Who Benefits from Benefits?
Municipal Consequences of Federal Cash Transfers

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Abstract

Centralized welfare programs limit local discretion over eligibility and payments, yet access often depends on municipal bureaucracies. We study whether local politicians obtain electoral returns from a federally designed cash transfer when their remaining discretion lies in registration. In Brazil's *Bolsa Família*, a 2009 reform unexpectedly changed the formula assigning municipal beneficiary quotas, and a later federal revision opened additional slots during the same mayoral term. We measure the cumulative authorization change relative to the allocation implied by the previous rule. In the 2,078 municipalities where the mayor elected in 2008 sought reelection in 2012, a one-standard-deviation quota shock created 171.50 additional authorized slots per 10,000 residents and induced 54.68 additional beneficiary families per 10,000 residents. The 2SLS estimate implies that 100 additional beneficiary families per 10,000 residents raise vote share by 2.59 percentage points. Enrollment expanded both with and without a pre-existing local welfare center (*CRAS*), but we find a clear electoral response only in *CRAS* municipalities. Within that group, the vote-share response is larger where offices report active outreach and more intensive service to *Bolsa Família* families. We interpret these patterns as scope conditions for the local implementation channel. Centralizing eligibility and payment does not eliminate local political returns when program access still depends on municipal implementation.

Keywords: cash transfers; electoral returns; state capacity; *Bolsa Família*.
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1 Introduction

Across the developing world, cash transfer programs have become a leading tool for reducing poverty while insulating redistribution from local political discretion. By standardizing eligibility rules, fixing benefit values by federal decree, and processing payments through centralized banking systems, these programs are designed to reduce the scope for patronage and discretionary allocation (Fiszbein and Schady, 2009). Yet access often depends on local bureaucracies controlled by elected officials. Can local politicians benefit electorally from policies designed and financed by another level of government? This is the question we address.

We study this question in the context of Brazil's *Bolsa Família*, one of the largest conditional cash transfer programs in the world.[1] Although eligibility rules, benefit size, and payments are determined by the federal government, municipalities remain responsible for identifying and registering eligible households, mostly through local welfare centers known as *Centros de Referência de Assistência Social* (*CRAS*). This institutional arrangement creates a narrow but important margin of local discretion: while mayors cannot alter transfer amounts, redirect funds, or withhold payments, they control the administrative pipeline through which households may gain access to the program.

This margin depended on a registration process that required active municipal effort. Identifying potentially eligible families meant locating them, conducting household interviews, and collecting documentation; municipal authorities controlled the local registration apparatus, which gave them influence over the pace, intensity, and reach of enrollment efforts.[2] In practice, two municipalities facing similar levels of underlying need could generate very different realized coverage depending on how actively the local administration pursued registration.

Our empirical strategy starts from a 2009 reform that revised the formula used to compute municipal *Bolsa Família* quotas. The program operates through federally determined municipal quotas, which set a ceiling on the number of families that can be served in each municipality.[3] A mayor elected in October 2008 could expect the

[1] By June 2006, *Bolsa Família* covered 11.10 million families, approximately 46 million people, or about 25% of the Brazilian population. By June 2015, the program supported 13.80 million families, with an average monthly benefit of R$167.15 and a budget equivalent to 0.50% of GDP.

[2] Hellmann (2015) describes enrollment as a municipal responsibility. He reports that as of September 2014, 77% of registrations were made without a home visit. Municipalities were responsible for preparing the local structure for registration, training personnel, promoting awareness campaigns, and designing *active search* strategies to reach eligible families outside the existing registry.

[3] Lindert et al. (2007) explains that *Bolsa Família* maintained municipal program quotas based on local

next adjustment to follow the previous allocation rule. Instead, the federal government changed that rule in 2009 and revised municipal quotas again in 2012.[4] These revisions expanded the ceiling, not the registry. Families who were already registered could enter when new slots opened, while filling the remaining slots depended on local governments' effort to identify, locate, and register eligible families not yet in the federal database.

We combine administrative data on quotas, realized beneficiary counts, and local social-assistance infrastructure with municipal electoral data from the subsequent election. The construction of the 2009 benchmark follows Gerard et al. (2025): we apply the previous allocation rule to the information available before the reform. Our main measure subtracts that benchmark from the official 2012 quota, capturing the cumulative authorization that opened during the mayor's term. We use this quota change to instrument realized beneficiary expansion.

Our findings trace the reform along four stages: the relationship between federal authorization and realized enrollment; the electoral response to the authorization shock; the 2SLS estimate that combines the two; and the institutional conditions under which these patterns emerge.

The first stage quantifies how federal authorization translated into realized enrollment. A one-standard-deviation increase in the quota shock corresponds to 171.50 additional authorized beneficiary-family slots per 10,000 residents. It raises realized beneficiary expansion by 0.30 standard deviations, equivalent to 54.68 additional beneficiary families per 10,000 residents.

Once federal authorization is shown to translate into realized enrollment, we estimate the electoral return to expanding the ceiling itself, before separating out how much of that ceiling municipalities chose to fill. For the same one-standard-deviation quota shock, the reduced-form intention-to-treat estimate is 1.41 percentage points on the incumbent's vote share. This is the electoral consequence of additional federal authorization, integrating over the local enrollment response documented in the first stage.

Scaling the intention-to-treat effect by the first stage gives the electoral response

poverty estimates. Original municipal allocations were constructed by combining the 2001 PNAD with the 2000 Census, generating an initial national target of 11.20 million families when the program was created in 2003.

[4] Hellmann (2015) notes that *Bolsa Família* administered municipal quotas using estimates of poor families developed in partnership with IBGE and reports that, in 2009, the federal government refined the measurement of the target population by incorporating methodologies to capture income volatility among vulnerable households. These studies formed a major expansion of coverage, incorporating more than 1.60 million additional families.

to realized beneficiary expansion. A one-standard-deviation increase in instrumented enrollment, about 181.36 additional beneficiary families per 10,000 residents, raises incumbent vote share by 4.69 percentage points. Equivalently, an increase of 100 beneficiary families per 10,000 residents raises vote share by 2.59 percentage points. The estimate is similar across alternative quota definitions, outcome definitions, and corrections for selection into the rerunning sample.

These returns differ across the institutional layer that converts authorization into delivery. Of the 2,078 municipalities in the estimation sample, 1,394 had a pre-existing local welfare center in 2008 and 684 did not. Realized enrollment expands in both groups: the first-stage coefficients are 0.30 with a *CRAS* and 0.32 without one. The divergence is electoral. The corresponding 2SLS estimate is 6.11 in the *CRAS* group; outside it, the confidence set includes zero. These estimates locate a clearer electoral response in municipalities with established offices, although balancing observed municipal characteristics substantially reduces that contrast.

Within these municipalities, we next examine two measures of how the local welfare apparatus operated. The first records whether municipal teams proactively searched for potentially eligible families (active outreach) rather than waiting for them to appear at a registration point. The first stage remains positive both with and without this practice, while the reduced-form vote-share point estimate is 2.00 percentage points with active outreach and 0.76 without it. The second measure records whether the *CRAS* reported serving *Bolsa Família* families through PAIF (*Serviço de Proteção e Atendimento Integral à Família*), a continuing family-support service provided through the social-assistance system. The corresponding reduced-form estimates are 2.25 percentage points where at least one *Bolsa Família* family was served through PAIF and 0.65 where none was reported. The PAIF contrast becomes sharper when we consider service volume: among municipalities reporting at least one family served through PAIF, the reduced form is 5.36 above the median number of families served and indistinguishable from zero below it. Because both operational measures predate the quota shock but were not randomly assigned, we interpret these results as scope patterns. They motivate a contact interpretation, but subsequent tests of operating duration and reported list-use frequency do not establish a consistent electoral gradient. Comparisons across office-opening cohorts are descriptive, and the evidence leaves the specific CRAS mechanism unresolved.

These findings advance three related literatures. We speak first to the political economy of cash transfer programs. Studies of social transfers find gains in government support and, in some settings, incumbent voting, although electoral results vary across de-

signs and contexts (Manacorda et al., 2011; De La O, 2013; Imai et al., 2020; Frey, 2019). Much of this literature studies settings in which local discretion is deliberately minimized, relying on centrally determined thresholds, randomized assignment, or other exogenous variation to identify electoral effects. We complement this work by studying a setting in which the program architecture is centralized but the last mile of delivery is not. Our findings suggest that centralizing transfer rules and payment flows is not sufficient to depoliticize redistribution when access to the program depends on a locally controlled registration bureaucracy. Even when mayors cannot alter eligibility criteria or benefit levels, control over the administrative pipeline through which households gain program access can still generate measurable electoral returns.

Our paper is especially closely related to Frey (2019), but the two designs study electoral returns under different implementation regimes: Frey's intervention leaves less room for mayoral involvement, whereas ours depends directly on municipal administration. Frey exploits a discontinuity in federal funding for the *Family Health Program*, a primary-care program in which teams follow a federally specified protocol of routine home visits to poor households. His information channel is embedded in those visits: the teams disseminate *Bolsa Família* eligibility information as part of their routine work, making enrollment less dependent on a separate effort by the mayor's welfare administration. We study the complementary margin. In our setting, converting expanded federal authorization into realized enrollment requires municipal welfare staff to identify, locate, and register eligible families. The designs also draw variation from different parts of the municipal distribution. Frey's threshold design is concentrated among small, low-development municipalities near a funding threshold, whereas the quota reform created more room for expansion in larger municipalities that had not yet absorbed earlier increases. Within his sample, coverage growth through a federally structured outreach channel reduces incumbency advantage and shifts the composition of challengers toward less clientelistic parties. Within ours, coverage growth through a locally managed registration channel produces positive returns for the incumbent.

Second, we contribute to the literature on political accountability under shared governance. A central question in this literature is whether voters can attribute responsibility when multiple levels of government jointly produce policy outcomes (Przeworski et al., 1999; Besley, 2006). Ferraz and Finan (2008) show that voters punish corrupt mayors when randomly assigned audits reveal misuse of federal funds, especially where local media facilitate information transmission. Avis et al. (2018) show that audits also reduce corruption through a legal disciplining channel. Our setting differs because the

relevant outcome is positive delivery rather than malfeasance. The results are consistent with voters giving local incumbents some credit for the expansion of a federally funded program when access still depends on municipal action. We do not observe voters' beliefs directly, but the pattern fits an attribution channel in a setting of shared responsibility.

Third, we contribute to the literature on state capacity and the bureaucratic foundations of policy delivery. Besley and Persson (2009, 2011) establish state capacity as a precondition for credible policy delivery, and Finan et al. (2017) survey the personnel and bureaucratic infrastructure through which public programs are implemented. Our findings distinguish enrollment pass-through from political conversion. Municipalities both with and without a pre-existing *CRAS* translate the quota shock into additional beneficiaries. The initial electoral contrast favors municipalities where the delivery apparatus was already present, but is sensitive to their observed composition. This distinguishes the evidence for local electoral returns from the more demanding claim that welfare offices themselves produce those returns.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting of *Bolsa Família*, the quota system, the federal revisions, and the data sources. Section 3 presents the impacts of the reform and summarizes the identification strategy, while the full empirical strategy appears in Online Appendix A and the figures and tables are collected after the references. Section 4 examines local implementation as a possible mechanism, evaluates the institutional comparisons, and discusses alternative explanations. Section 5 concludes.

2 Institutional Background and Data

2.1 *Bolsa Família* Program

Program Design. Launched in 2004, *Bolsa Família* consolidated several pre-existing social-assistance programs into a single conditional cash transfer system and became Brazil's flagship poverty-alleviation policy. In 2009, the program classified eligible families into two income groups. Families with monthly per capita income at or below R$70 were classified as extremely poor and received a basic monthly benefit of R$68, plus variable benefits of R$22 per child or adolescent aged 0–15, up to three, and R$33 per adolescent aged 16–17, up to two. Families with monthly per capita income above R$70 and at or below R$140 qualified only for the child- and adolescent-linked variable benefits. These benefits were conditional on age-specific education and health require-

ments, and the maximum monthly transfer per family was R$200. Table 1 summarizes the program's eligibility thresholds and benefit values from 2004 to 2018. By December 2009, the program reached approximately 12.40 million families, with an average benefit of about R$95.

Federal Rules and Local Responsibility. *Bolsa Família* combined centralized rules and payments with decentralized administration. The federal government, through the *Ministério do Desenvolvimento Social* (MDS), set eligibility criteria and benefit values, used *Cadastro Único* records to identify families eligible for payment, and transferred benefits directly through the state-owned bank *Caixa Econômica Federal*. State governments coordinated implementation across social assistance, education, and health and provided technical support to municipalities, but neither selected beneficiaries nor disbursed transfers.[5]

Municipal governments operated the program locally. They identified low-income families, collected and updated their registry information, conducted household visits, monitored compliance with education and health conditions, and coordinated the municipal agencies involved in implementation. Mayors therefore could not change eligibility thresholds or benefit values, redirect federal transfers, or guarantee that a registered family would receive a benefit. Their administrations could, however, affect how intensively potential beneficiaries were sought, how quickly applications were processed, and how completely eligible families became visible to the federal system. This separation between federal benefit rules and local implementation is the institutional margin central to our analysis.

Municipal Registration. The *Cadastro Único* (*CadÚnico*) is a national registry of low-income families and the administrative gateway to *Bolsa Família* and many other social programs. It covered families with per capita income below half the minimum wage (R$232.50 in 2009), well above the program's own eligibility thresholds. Registration was therefore necessary but not sufficient for benefit receipt, and the registered population was substantially larger than the beneficiary population. Under Decree No. 6,135/2007, the MDS defined and supervised the national registry while participating municipalities collected family information using a standardized form.

Registration required a face-to-face interview covering household composition,

[5]The division of responsibilities during the study period was governed principally by Law No. 10,836/2004, Decree No. 5,209/2004, and MDS Ordinance No. 246/2005. The ordinance formalized municipal adherence to the program and specified federal and municipal commitments.

housing conditions, assets, employment, and income. In urban areas, families could seek a municipal registration post, often located at a *Centro de Referência de Assistência Social* (*CRAS*). In rural and remote areas, reaching a fixed office could be costly or impractical. Coverage there depended more heavily on active outreach: municipal teams had to locate potential beneficiaries, travel to underserved communities, and conduct interviews in or near their homes. Federal rules remained uniform, but the reach and timing of this local effort shaped which families entered the registry and when.[6]

2.2 Institutional Details of the Quota Reform

Quotas and Enrollment. Eligibility alone does not guarantee benefit receipt. Access to *Bolsa Família* depends on the availability of municipal quotas: caps on the number of beneficiaries per municipality, defined nationally and revised approximately every three years based on poverty estimates derived from census and survey data. These quotas function as official upper bounds on municipal enrollment, but the constraint is not always immediately binding. In practice, municipalities frequently operate near their ceiling, oscillating modestly above or below it from month to month before the federal priority mechanism reallocates unused slots.

Priority and Local Effort. The priority index considers both quota utilization and unmet registered demand: municipalities with low beneficiary-to-quota ratios and few eligible-but-unserved families risk seeing unused slots reallocated to places with stronger registered demand. The quota system therefore creates pressure to use the available slots. A quota increase is not a benefit guarantee, but it opens federally authorized room that municipalities must fill by identifying, registering, and updating eligible households. Effective coverage thus depends on local administrative initiative: political will and organizational capacity determine how much of the authorized ceiling becomes actual benefit receipt.

Quota Reform. In 2009, the federal government revised the methodology used to compute municipal quotas, producing a heterogeneous reallocation of authorized ben-

[6] A *CRAS* is the local public facility responsible for basic social-assistance services and a visible point of contact between families and municipal government; it is not the only place where *CadÚnico* interviews may occur. The 2009 *Censo SUAS* recorded 5,798 units in 4,329 municipalities. Municipalities without a *CRAS* could still register families through other municipal facilities or outreach teams, so absence of a unit does not imply absence of registration capacity.

eficiary slots across municipalities. The previous approach, in place since 2006, distributed slots in proportion to each municipality's estimated number of poor families from the 2000 Census, adjusted for intercensal population growth. The 2009 revised methodology accounts for income volatility among families near the eligibility line, a feature of transient poverty that earlier projections systematically undercount (Gerard et al., 2025).

Reallocation Shock. Alongside the methodological update, all municipal quotas were scaled by a factor of 1.18 to account for structural income volatility near the poverty threshold. Because this multiplicative factor is constant, it contributes no cross-municipal variation to the instrument; all identifying variation in the quota shock derives from the updated municipal poverty estimates and the change in income thresholds. Together, these changes added nearly 1.60 million new authorized slots in 2009. The reform expanded the ceiling, not the registry: although already-registered eligible families were absorbed automatically, the magnitude of the expansion left a large residual margin of unfilled slots whose realization depended on the municipal government's effort to identify, locate, and register eligible families not yet in the federal database.

2.3 Data

Our empirical analysis combines five sources of data: administrative records on *Bolsa Família*, family-level records from the *Cadastro Único*, electoral returns, census and household-survey data, and administrative records on local social-assistance infrastructure.

***Bolsa Família* Administrative Records.** We begin with monthly administrative records from the *Ministério do Desenvolvimento Social* on *Bolsa Família* quotas, realized coverage, and total transfers from 2006 to 2012. These data span the pre-reform baseline, the 2009 reallocation, and the post-reform expansion window through the 2012 election, and allow us to measure both authorized and realized program expansion.

***Cadastro Único*.** We also use family-level records from the 2012 *Cadastro Único*, provided by the *Ministério do Desenvolvimento Social*. The file covers families recorded in the registry in 2012, including those registered in earlier years. It reports their municipality of residence, per-capita family income, family characteristics, and dates of first

registration and last update. The first-registration date identifies entry into the registry separately from subsequent updates. We use it to construct registration cohorts, including families first registered before the reform.

Snapshot and Updating. Income and family characteristics reflect the information on file in 2012. During the study period, registry information required updating or revalidation every two years.[7] A family first registered in 2007 can therefore be assigned to its registration cohort using the 2012 file. Its recorded income, however, describes its circumstances as last reported by 2012. The cohort comparisons describe families observed in that snapshot, grouped by first-registration date and income recorded in 2012. Families absent from the snapshot and their earlier income histories remain outside this exercise.

Electoral Records. We then use electoral records from the *Tribunal Superior Eleitoral* (*TSE*) for the 2008 and 2012 municipal elections. These data provide candidate-level vote totals, turnout, and detailed candidate characteristics, including education, prior occupation, and party affiliation.

Census and Household Surveys. To reconstruct counterfactual municipal quotas under the pre-reform allocation rule, we combine the 2000 Census from the *Instituto Brasileiro de Geografia e Estatística* (*IBGE*) with poverty estimates from the 2006 *Pesquisa Nacional por Amostra de Domicílios* (*PNAD*). These are the inputs used to approximate the quota formula that would have prevailed without the 2009 reform.

Censo SUAS. Finally, we use administrative records from the *Censo SUAS* covering *CRAS* units in 2008 and 2009. These data describe local infrastructure, active outreach, and recorded numbers of beneficiary families served. We use the family count as a provisional proxy for PAIF activity, subject to the measurement limits discussed in Section 4.3. To examine office duration and network expansion, we also harmonize the annual censuses from 2007 through 2012. Absence from a census is treated as missing information rather than evidence that an office closed.

[7] Article 7 of Decree No. 6,135 of June 26, 2007 established the two-year validity of registry information.

2.4 Sample Construction

Municipal Panel and Data Availability. Our electoral analysis uses municipality-year observations for the 2008 and 2012 elections. We combine the sources using municipal identifiers and require observations in both years. The integrated database contains 5,568 municipal identifiers across election years. This is the starting universe for the sample accounting in Table C.2. Requiring the inputs for quota shocks and realized enrollment excludes 63 municipalities, leaving 5,505. Missing records identify gaps in the measures needed for estimation. They do not establish that a municipality lacked access to *Bolsa Família*. Program exposure is measured in beneficiary families: the quota shock changes authorized family slots, and realized expansion measures the net change in beneficiary families at the municipal level.

Electoral Sample and Record Quality. We follow the mayor elected in 2008 and measure that mayor's vote share in each election. Requiring a candidacy in 2012 excludes 3,140 municipalities from the preceding universe and leaves 2,365. We then exclude 1 municipalities with incomplete electoral information and 83 municipalities with an uncontested election in either year. A further 84 municipalities lack a positive recorded vote share for the tracked mayor. Linking candidates across elections also requires an unambiguous name match, which excludes 7 municipalities. Removing 85 municipalities with multiple elected-status records leaves 2,105. We additionally check turnout identities and election validity. These checks impose no further losses in the displayed sequence. Table C.2 reports every restriction, including those with no incremental exclusions.

Final Estimation Sample and Selection. Complete information on municipal transfers excludes 24 municipalities, and missing mayor characteristics exclude another 3. The final sample contains 2,078 municipalities and 4,156 municipality-year observations. We use this common sample across the main specifications. Its construction also requires the variables used in the alternative windows and mechanism comparisons. Table 2 compares it with a broader descriptive sample of 4,976 municipalities. The selection diagnostic uses a separate complete-case universe of 5,492 municipalities before conditioning on candidacy.

Selection into Rerunning. Selection matters because the reform could affect whether the incumbent runs again. In the preferred selection specification, a one-standard-

deviation quota shock predicts a 0.04 percentage-point change in that probability. Reweighting the electoral regression by the inverse probability of selection yields an estimate of 4.14 percentage points, compared with 4.69 in the main specification. These exercises provide evidence on selection through observed characteristics, while selection on unobserved determinants remains a limitation. Online Appendix A, especially Section A.3.1, presents the selection tests and their uncertainty.

2.5 Descriptive Statistics

Municipality and Mayor Characteristics. Table 2 describes municipalities and their elected mayors in the broader comparison sample and the estimation sample. The 2008 winner's vote share averages 54.97 percent in the former and 52.92 percent in the latter. Predetermined municipal characteristics, including urbanization, poverty, income, and inequality, have similar means across the two panels. Mayor characteristics differ more, including prior occupation and age. The estimation sample describes incumbents who contest another election, so these differences help delimit the population studied. Similar municipal averages alone do not establish that selection is unrelated to electoral outcomes.

Program Coverage and Expansion. Municipalities in the estimation sample had an average of 879.87 beneficiary families per 10,000 residents in 2008. Realized coverage rises by 203.50 families per 10,000 residents on average through 2012, with a median increase of 163.15. The realized measures use twelve-month averages of monthly beneficiary-family counts. Rates use municipal population in 2006 as a fixed denominator, so changes reflect enrollment rather than contemporaneous population growth. Authorized and realized expansion vary across municipalities and are right-skewed. We express regression coefficients per one analysis-weighted standard deviation to make their scale interpretable and translate the estimates into families per 10,000 residents. These measures count families receiving benefits, whereas the *Cadastro Único* cohort exercise describes registered families using the 2012 snapshot.

Local Social-Assistance Infrastructure. The estimation sample includes 1,394 municipalities with a pre-existing *CRAS* and 684 without one. Municipalities with a *CRAS* account for 67.08 percent of the sample. The infrastructure measures also distinguish whether these units conduct active outreach and provide family-assistance services. These characteristics describe the local capacity through which eligible families may en-

ter the registry and obtain benefits. Section 4 examines whether the electoral response to expansion differs across these settings. That comparison conditions on existing infrastructure and does not identify the effect of installing a *CRAS*.

Figures 1, 2, and 3 collected after the references complement these statistics with the evolution of quota utilization, the relationship between authorized and realized enrollment, and the distribution of unused authorization. The tables discussed below are collected there as well.

3 Impact of the 2009 *Bolsa Família* Reform

Municipal incumbents gain an estimated 4.69 percentage points of vote share following a one-standard-deviation expansion in local enrollment induced by federal authorization (Anderson–Rubin 95 percent set: [1.24, 10.28]). We begin by documenting how federal authorization changed municipalities' room to enroll families. We then ask whether that authorization translated into electoral support, and whether the answer depends on realized enrollment. The federal government sets the program's quotas and pays its benefits, while municipal offices identify and register eligible families. Our estimates concern mayors who sought reelection and the enrollment expansion induced by federal authorization. The main tables and figures appear at the end of the text, before the appendix.

3.1 Policy-Induced Variation in Authorization and Coverage

Authorization. The reform created variation in how many families municipalities were authorized to enroll during the mayoral term. The 2009 reform changed that authorization, and further quota revisions followed in 2012. We measure cumulative authorization as the official 2012 quota minus the counterfactual quota that the previous allocation rule would have implied in 2009, evaluated at the R$90/137 poverty thresholds. The comparison holds the earlier rule as the benchmark and follows enrollment through the end of the term. One standard deviation of the authorization shock corresponds to 171.50 additional authorized families per 10,000 residents.

Realized Enrollment. Did authorized capacity translate into realized coverage? Municipalities enrolled more families where the federal government authorized greater expansion, across the reported control sets. An increase of one standard deviation in

authorization brings an estimated 54.68 more beneficiary families per 10,000 residents (Table 3). Each additional authorized place corresponds to 0.32 additional families receiving the benefit.[8] This is where municipal implementation enters: federal authorization opens places, while local staff identify and register eligible families who may fill them.

3.2 Identification Strategy

Mayors may expand enrollment in response to electoral pressure, and municipalities differ in their ability to reach eligible families. A direct comparison of enrollment growth and votes would combine these local choices with the political consequences of the program. We instead compare changes in incumbent support between municipalities exposed to different federal authorization shocks, conditional on pre-reform characteristics and incumbent party.

Let $\mathbf{D}$ collect municipalities' enrollment changes, $\mathbf{Z}$ their authorization shocks, and $\Delta\mathbf{Y}$ their changes in incumbent vote share. The matrix $\mathbf{X}^0$ contains pre-reform controls, including incumbent party. We estimate

$$\mathbf{D} = \pi_0\mathbf{1} + \pi_1\mathbf{Z} + \mathbf{X}^0\boldsymbol{\pi}_2 + \boldsymbol{\eta}, \tag{1}$$

$$\Delta\mathbf{Y} = \beta_0\mathbf{1} + \beta_1\widehat{\mathbf{D}} + \mathbf{X}^0\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}, \tag{2}$$

where $\widehat{\mathbf{D}}$ contains the first-stage fitted values and $\mathbf{1}$ is a vector of ones. We estimate turnout in the same way. Enrollment and authorization are rates per 10,000 residents using fixed 2006 population, divided by their respective analysis-weighted standard deviations on the common sample. Electoral responses are reported in percentage points. Throughout, intervals accompanying IV estimates are Anderson–Rubin 95 percent sets.

We weight municipalities by 2006 population, so the estimates give greater influence to places where more people live. Standard errors are clustered by state. With two periods and fixed municipality weights, equations (1) and (2) give the same point estimates as the municipality and year fixed-effects panel reported in the tables.

Validity of the Design. The authorization comparison requires the quota shock to be unrelated to the electoral trend that would otherwise have occurred, conditional on the

[8] On the standardized scale, the first-stage coefficients are 0.32, 0.36, and 0.30 as the control set expands. Enrollment growth compares the twelve-month average family counts in 2008 and 2012, allowing for entries and exits. These coefficients describe net expansion, not an individual family's probability of receiving a benefit.

controls. Interpreting the enrollment coefficient additionally requires authorization to affect votes through realized enrollment and a continuous-treatment monotonicity condition. Online Appendix A develops these assumptions and examines selection into rerunning, pre-reform electoral balance, alternative exclusion channels, and first-stage monotonicity. The diagnostics help assess the design; they cannot establish exclusion or eliminate selection through unobserved characteristics.

3.3 Electoral Responses

Authorization and Incumbent Vote Share. The authorization comparison asks whether mayors receive more votes where the federal government permits greater expansion. The preferred reduced-form estimate is 1.41 percentage points per standard deviation of the quota shock (Table 4). With the preceding control sets, the estimates are 1.93 and 1.30. This comparison captures the electoral response to authorization, including any response that operates before families actually receive benefits.

Realized Enrollment and Incumbent Vote Share. The electoral return to coverage is 4.69 percentage points of incumbent vote share for a one-standard-deviation expansion in local enrollment ([1.24, 10.28]; Table 5). The estimated electoral gain can be distinguished from zero, although its magnitude remains uncertain. To put the change in terms of families served, an additional 100 beneficiary families per 10,000 residents corresponds to an estimated 2.59 percentage points of vote share ([0.68, 5.67]).[9] For comparison, the median incumbent lost 3.03 percentage points of vote share between these elections. The estimate describes a change in the mayor's electoral support at the municipal level; it does not reveal which voters changed their vote or how they assigned credit.

Turnout. Turnout provides a separate margin of electoral response. We do not find a clear turnout response to local program expansion; the confidence set includes zero. The preferred response to authorization is -0.35 percentage points, and the response to instrumented enrollment is -1.17 ([-2.79, 0.05]). Tables 4 and 5 report both outcomes under the same control sequences. Separating turnout from incumbent vote share matters because receiving more votes and bringing more voters to the polls are distinct political responses.

[9] One standard deviation of enrollment growth is 181.36 beneficiary families per 10,000 residents. The natural-unit coefficient and interval rescale the same IV estimate.

3.4 Heterogeneity and Scope

Municipality Size. The pooled response is concentrated in municipalities of a particular size. The electoral gains are clearest among mayors of larger municipalities. The confidence sets exclude zero in the two upper population tails, while the evidence for the two lower-population samples remains inconclusive. Population weighting makes this distinction relevant to the interpretation of the pooled result. Table 6 reports estimates for municipalities below the median, below the 75th percentile, and in the two upper population tails.[10] These comparisons show where the estimated response appears in the sample. They do not identify the effect of making a municipality larger.

Baseline Coverage and Prior Expansion. Prior coverage may leave less room for the authorization shock to operate. The conditional estimates point to electoral gains below median baseline coverage and remain inconclusive above the median. Below and above median baseline coverage, the estimates are 10.19 ([1.72, 20.80]) and -0.41 ([-1.84, 0.81]), respectively (Table 7). These paired comparisons use conditional inference, which limits what we can conclude about differences between the groups.[11]

The same table compares municipalities inside and outside the Frey (2019) sample. The outside and inside estimates are 14.20 ([3.87, 52.46]) and -0.12 ([-1.71, 1.48]). The conditional estimates point to electoral gains outside the Frey sample and remain inconclusive inside that sample. The conditional first-stage F statistics are 9.75 and 44.76, respectively. This comparison relies on the enrollment response within each group. Earlier expansion could have reduced the eligible pool still available for enrollment. But coverage in 2008, although measured before our shock, may reflect that earlier expansion as well as other municipal differences. The comparison cannot establish that the program had exhausted the eligible pool or that saturation explains the electoral pattern.

Region and Party. The regional and partisan comparisons locate the pooled return across municipalities and incumbents. Outside and inside the Northeast, the estimates are 2.71 ([-1.79, 9.13]) and 1.01 ([-0.73, 2.60]; Table 8). The regional comparison does

[10] Below the median and below the 75th percentile, the IV estimates are 0.07 ([-1.44, 1.37]) and 0.11 ([-1.47, 1.79]). Above the 75th percentile, the reduced form is 2.91 and the IV estimate is 11.27 ([3.37, 27.32]). Above the 90th percentile, they are 5.42 and 18.26 ([8.68, 60.48]).

[11] The baseline-coverage, Frey, region, and party partitions allow instruments, endogenous variables, and the complete control specification to differ across groups. The joint regression retains their covariance. The reported AR sets hold the other group's coefficient at its estimate; they are conditional diagnostics rather than joint confidence regions robust to weak identification of both coefficients. Differences use conventional state-clustered tests and are not robust to weak instruments.

not establish a difference in the electoral return under the conventional state-clustered test at the five-percent level. For non-PT and PT incumbents, the estimates are 6.49 ([1.67, 13.10]) and -1.37 ([-10.59, 13.50]). The conditional estimates point to electoral gains for non-PT incumbents and remain inconclusive for PT incumbents. The party-group comparison does not establish a difference in the electoral return under the conventional state-clustered test at the five-percent level. These comparisons describe the reach of the pooled finding; they do not identify how voters divide credit between federal and local politicians.

3.5 Robustness

Outcome Definitions and Extreme Values. The electoral return could depend on how we measure support or on a few extreme observations. Table 9 addresses both concerns. Winsorizing vote share gives an estimate of 4.23 ([1.16, 9.21]), while using registered voters as the denominator gives 3.80 ([0.91, 8.41]). Winsorizing the authorization shock gives 5.50 ([1.78, 11.01]). The registered-voter measure counts support relative to the electorate, rather than only voters who cast a valid vote for a candidate.

Authorization Horizon. The 2012 quota revision may matter for the expansion observed over the mayoral term. Using only the official 2009 quota relative to the counterfactual allocation gives an estimate of 2.47 ([-3.25, 7.85]). Enrollment still measures the full 2008–2012 change. This comparison asks whether the electoral result depends on the later authorization endpoint; it does not isolate the return to families enrolled during 2008–2009.

Selection into Rerunning. The sample contains mayors who chose to seek reelection. Reweighting for that choice using observed characteristics gives an estimate of 4.14 ([0.98, 8.81]). This adjustment combines inverse-probability and population weights. Its value depends on overlap and on the selection model, which cannot account for unobserved determinants of both rerunning and electoral performance.

Normalization and Discretization. We also examine whether the result depends on the way municipal expansion is measured. Using Census-2000 families as the denominator for enrollment and authorization gives 4.60 ([0.77, 10.91]). This changes the definition of exposure while retaining population weights in estimation.

A separate exercise asks whether authorization moves municipalities across successively higher thresholds of enrollment growth (Table 10). The estimates for crossing the median, 75th-percentile, and 90th-percentile thresholds are 21.95 ([5.09, 67.37]), 17.65 ([5.44, 33.66]), and 22.45 ([6.67, 47.81]). The continuous specification gives 4.69 ([1.24, 10.28]). Because every panel keeps the same authorization instrument, the reduced form remains 1.41 by construction. The threshold estimates refer to different changes in program coverage than the continuous coefficient. Their causal interpretation requires exclusion through the chosen binary treatment, so they do not by themselves recover an unrestricted dose-response relationship.

3.6 What Expanded Access Means for Families

The electoral return accompanies an expansion in access to a program intended to improve families' living conditions. Registration can open a path to assistance, although entering *Cadastro Único* does not guarantee a transfer. For families who become beneficiaries, access brings income support and links to health and education services. Existing research helps explain why this access may matter. Appendix B reviews that evidence study by study; it provides context for our findings rather than estimates of the household effects of this reform.

Health. Rasella et al. (2013) document lower childhood mortality as *Bolsa Família* coverage expands, particularly from poverty-related causes. Their findings suggest that access to the program can have consequences beyond the transfer itself. Our municipal electoral estimates do not measure those health gains or distinguish the contribution of income from that of health services.

Education. Glewwe and Kassouf (2012) find higher school enrollment, lower dropout rates, and greater grade promotion following the introduction of *Bolsa Escola/Bolsa Família*. These outcomes give substance to the value of reaching eligible families with children. They concern schooling under the broader program, however, and do not establish the educational effect of the additional registrations induced by the quota reform.

Income and Poverty. Using household surveys, Ferreira de Souza et al. (2019) show that program transfers reduce measured poverty and inequality when included in household income. This accounting evidence illustrates the immediate importance of

benefit receipt for poor families. Taken together, the literature makes it plausible that expanded access yields material gains alongside the mayor's electoral return. It does not tell us how much of either gain depends on new registration, or whether families reward the mayor for the income, the assistance in obtaining it, or both. We turn next to the municipal institutions through which families encounter the program.

4 Possible Mechanisms

Why might mayors gain votes when the federal government expands a transfer program? Helping families register may make the municipality's contribution visible, while accompanying services may give families additional reasons to support the incumbent. Income gains could also change political preferences or participation without families crediting the mayor for access to the program. Our data do not fully distinguish these explanations. We examine pre-existing welfare offices and their activities to assess whether contact with municipal staff helps account for the electoral gains, then consider the alternatives.

Families receive *Bolsa Família* payments from the federal government, but often encounter the program through municipal staff. Help with registration may lead families to associate a federal benefit with the local administration. The transfer literature also points to beliefs about government and reciprocity as possible sources of political support, without isolating the role of contact with program staff (Manacorda et al., 2011).

We use the quota-induced expansion from Section 3 to compare enrollment and electoral gains across municipalities with different welfare infrastructure.[12] Because offices and service practices were not randomly assigned, these comparisons can inform the contact explanation without identifying the effect of providing those services.

4.1 Local Welfare Offices and Electoral Returns

Table 11 reports the electoral estimates using registration expansion as the endogenous variable. Quota expansion also increases first registrations in *Cadastro Único*. Table F.8

[12]The subgroup IV comparisons start from the main sample and retain its scaling, population weights and baseline controls, with further sample restrictions stated below, and allow the instrument, treatment and controls to vary by group. Standard errors are clustered by state. Group-specific Anderson–Rubin sets are conditional diagnostics that hold the other group's coefficient at its estimate; they are not joint confidence regions robust to weak instruments. Conventional difference tests accompany these comparisons. Additional reduced-form diagnostics report state-clustered intervals, wild-cluster tests and the specified multiplicity adjustments.

shows that the response is concentrated among families whose recorded income places them within the program's income eligibility range. This is consistent with municipalities bringing potential beneficiaries into the registry. The data are a snapshot taken after the reform: income is measured in that snapshot, and first registration does not establish subsequent benefit receipt or identify which families changed their votes.

A *CRAS* gives families a place to seek assistance, meet municipal staff, and obtain help accessing social programs. Of the 2,078 municipalities in our sample, 1,394 already had a recorded center in 2008 and 684 had none recorded. We ask whether mayors gain more votes from program expansion where families already have this point of contact with the administration.

Federal authorization expanded enrollment in municipalities both with and without a welfare center. The first-stage coefficients are 0.30 with a *CRAS* and 0.32 without one. The data do not clearly distinguish the two enrollment responses. Table 12 reports both enrollment and electoral outcomes.[13]

The electoral gains are concentrated in municipalities that already had a welfare center. A one-standard-deviation increase in enrollment induced by the quota shock corresponds to 6.11 percentage points of incumbent vote share with a *CRAS*, compared with -1.23 without one. The authorization estimates follow the same comparison: 1.84 and -0.40 percentage points, respectively.[14]

This contrast directs attention to the municipality's role in families' access to the program. A welfare office may make that role more visible even when families in other municipalities also receive the federal benefit. The comparison is consistent with local contact helping voters associate access to the program with their mayor. Other explanations remain open: municipalities with a center may differ in the families reached by expansion, the services those families receive, or other features of local administration.

Municipal differences matter for this interpretation. Table 13 describes the two groups using the same variables as the main descriptive table. Table F.20 reweights municipalities with and without a baseline office to balance observed characteristics. The difference in electoral returns becomes substantially smaller under those weights, while enrollment still increases in both groups (Table F.9). Reweighting changes which municipalities receive the most emphasis and leaves unobserved differences uncontrolled. The sensitivity cautions against attributing the original electoral difference to

[13]The test of equal first-stage coefficients gives $p = 0.682$. A failure to distinguish these coefficients does not establish that the enrollment responses are identical.

[14]The conditional AR sets for the vote-share estimates are [2.12, 13.45] with a *CRAS* and [-5.25, 2.99] without one. The conventional test of their difference gives $p = 0.007$.

the office itself.

4.2 Outreach and Family Contact

Outreach brings municipal staff to families who may not seek assistance at an office. It may make municipal effort visible to people who would otherwise have little contact with the welfare administration. Within municipalities that already had a *CRAS*, we compare those reporting active outreach before the reform with those without recorded outreach. The underlying coding groups missing responses with no recorded outreach, but an audit of the original source finds no missing outreach responses in this estimation sample.

The estimated electoral return is larger where offices report active outreach: 7.27 percentage points, compared with 1.74 without recorded active outreach. The estimates leave considerable uncertainty about the gap between these electoral returns. Table 14 reports the comparison.[15] The first-stage point estimate is smaller in municipalities reporting active outreach.

These offices were already conducting outreach before the reform. The comparison therefore cannot measure an outreach campaign caused by expansion. Their resources, the families they serve, or the visibility of local officials could also account for the difference in electoral returns.

Reported Contact Practices. If outreach helps families associate program access with the mayor, more frequent outreach might accompany larger electoral gains. The baseline census asks how often staff use lists to search for vulnerable families. We average the ordered responses across offices within each municipality and ask whether registration, enrollment and votes respond more strongly to quota expansion where this score is higher. Table F.10 reports the results within municipalities with a baseline CRAS. For a standard-deviation increase in the score, the electoral response to a standard-deviation quota shock increases by an estimated 0.394 percentage points [-0.691, 1.479]. The corresponding change in enrollment is 4.035 additional beneficiary families per 10,000 residents [-5.587, 13.656]. Both estimates are imprecise, and neither survives the stated multiplicity adjustment.

Table F.11 examines referrals to the registry, priority given to family follow-up, and an exploratory measure of outreach to families failing to meet program conditions. Ta-

[15] The conditional AR sets are [2.40, 13.66] with outreach and [-1.70, 7.41] without it. The conventional difference test gives $p = 0.051$.

ble F.12 uses an alternative coding of outreach frequency and allows quota responses to vary with other baseline municipal characteristics. These comparisons do not show a consistent link between reported contact practices and electoral gains. Referrals are almost universally reported, and population weights give relatively few municipalities much of the influence on the estimates. These staff reports measure neither completed registrations nor actual encounters or the credit voters assign to officials. The imprecision leaves room for meaningful contact effects.

4.3 Family Follow-Up through PAIF

PAIF is the family-support program delivered through CRAS, known before the reform as the *Programa de Atenção Integral à Família* (Comprehensive Family Support Program). It assists families facing social vulnerability and helps them access benefits and public services.[16] Follow-up may keep families in contact with municipal staff after they first seek assistance, giving them repeated opportunities to observe the municipality's role. We use the pre-reform record of *Bolsa Família* families served by welfare offices as a proxy for PAIF activity, although this count does not measure how often each family meets staff.

The estimated electoral return is larger where offices record families served through PAIF: 7.70 percentage points, compared with 1.76 without a positive recorded family count. The estimates leave considerable uncertainty about the gap between these electoral returns. Table 15 reports this comparison.[17] The first-stage point estimate is smaller in municipalities reporting PAIF follow-up.

We also distinguish municipalities by the recorded number of beneficiary families served, summed across their welfare offices. This comparison uses the 962 municipalities with a positive recorded count, divided at their pre-reform median; it therefore concerns a narrower population than the preceding comparison. The estimated electoral return is larger in the higher-volume group: 18.78 percentage points, compared with -1.51 in the lower-volume group. The evidence supports a difference in electoral returns between these groups. Table 16 presents the estimates.[18] The recorded family

[16]The Ministry's institutional history and description of PAIF dates its establishment to 2004. In 2009 it became the *Serviço de Proteção e Atendimento Integral à Família* (Comprehensive Family Protection and Support Service), retaining the acronym. Its activities include family support and referrals to the registry and other public services. The English names here describe the Portuguese titles.

[17]The conditional AR sets are [2.01, 15.78] with recorded PAIF follow-up and [-1.50, 9.53] without it. The conventional difference test gives $p = 0.095$. The conditional first-stage F statistics are 38.04 and 8.18, respectively.

[18]The conditional AR sets are [8.01, 35.05] in the higher-volume group and [-7.71, 5.34] in the lower-volume group. The conventional difference test gives $p = 0.017$. The lower group includes observations

count does not measure the quality of assistance or the amount of contact received by each family.

4.4 Institutional Exposure and Network Expansion

We harmonize the official CRAS censuses from 2007 through 2012, retaining both the office counts observed in each wave and the reported opening dates.[19]

An office that has operated for longer may give families more opportunities to encounter municipal staff. A recent opening could instead make local action especially visible. We measure years of operation using opening dates reported before the reform. Table F.13 asks whether quota expansion brings larger electoral gains where offices have operated for longer, allowing separately for whether a baseline office exists. The electoral response to a standard-deviation quota shock increases by an estimated 0.788 percentage points per additional year of operation [-0.049, 1.626], with adjusted $p = 0.952$. The estimate remains imprecise when we account for baseline office density or exclude dates that may refer to predecessor facilities. Table 17 compares offices established by 2006, those established in 2007–2008, and municipalities without a recorded baseline office.

The comparison between older and recent offices also depends on which municipalities receive the most weight. Within the baseline-CRAS sample, the older-minus-recent difference in the response to a standard-deviation quota shock is 1.444 percentage points [-0.154, 3.043] under population weights. After balancing the two groups on observed baseline characteristics, it is -0.436 [-2.036, 1.164] (Table F.14). Unobserved differences remain, and reweighting changes the population emphasized by the comparison. The apparent advantage of older offices does not persist under these weights.

Table 18 and Figure F.8 extend the comparison to offices reported as opening during the mayoral term. The late-entry group has a smaller electoral point estimate than the oldest baseline group: the difference per standard-deviation quota shock is -2.941 percentage points [-4.325, -1.558], with adjusted $p = 0.228$. The adjusted test does not establish this difference, and comparisons between successive opening cohorts do not

at the median. Both PAIF comparisons use a family-count item that lacks its complete original questionnaire label in our extract. Its interpretation as PAIF follow-up remains provisional. The binary comparison codes missing counts together with zeros. The count is a proxy for service activity, not a verified measure of repeated contact.

[19]The Ministry of Social Development provides the annual data, questionnaires and manuals through its Censo SUAS archive. Baseline duration uses dates reported in 2008, rather than dates recalled after the reform. Source questionnaires and manuals distinguish registry referral from registration and identify the list-use items used in the contact comparisons.

show that electoral gains rise systematically with years of exposure. These groups reflect post-reform decisions about office placement, which offices survive, and which have valid reported dates. Their comparison remains descriptive. Later adopters remain in the main estimation sample, and their entry does not redefine its control group.

Network Expansion and Identification. An office can first appear in the census without having just opened, and a new identifier need not mean a new office. We treat an unobserved municipality as missing and exclude reported openings when an earlier census already records that office in operation. Table F.15 checks whether municipalities with baseline offices continue to appear in later censuses. The adjusted tests do not establish that quota changes predict nonresponse. Missing census records still cannot verify closures or resolve uncertain opening dates.

Could quota expansion itself lead municipalities to open offices? This matters because new services might win votes independently of benefit receipt. Table F.1 tests whether quota changes predict the observed expansion of the network. For a standard-deviation increase in the initial 2009 shock, the estimated change in the probability of observing an office in 2012 among municipalities coded without a baseline CRAS is 0.289 percentage points, with a confidence interval of [-1.172, 1.750]. The estimated response of date-consistent reported openings is -0.062 offices [-0.189, 0.065]. These estimates do not establish that quota expansion led to more offices. One association in the broader set of tests survives adjustment: a standard-deviation increase in the cumulative quota shock is associated with a change of -0.212 offices by 2010 [-0.488, 0.064], with adjusted $p = 1.000$. This quota measure includes changes after the 2010 census, so the association cannot establish their causal effect on earlier openings. It nevertheless prevents us from concluding that quota changes are unrelated to infrastructure. Services that respond to authorization and independently affect votes would challenge the interpretation of the IV estimate as an effect operating only through enrollment.

4.5 Alternative Explanations

Material Benefits and Political Credit. Additional income could change how families evaluate government, independently of the office that helped them register. In Uruguay, Manacorda et al. (2011) find persistent government support after transfers end and discuss both beliefs about government and reciprocity as explanations. Our municipal election data do not reveal these beliefs or whom families credit for assistance. We also cannot separate broader local economic effects on voters who do not

receive transfers. Electoral gains could therefore arise without an encounter with municipal staff.

Participation and Electoral Composition. A mayor's vote share can rise because voters switch candidates, because turnout changes, or because new voters enter the electorate. Table 19 expresses incumbent votes, other candidates' votes, blank or null ballots, and abstention using a common electoral denominator. It also reports registered voters relative to fixed baseline population. Table 20 presents the corresponding 2SLS estimates. The registered electorate responds to quota exposure, while the CRAS differences in these outcomes do not survive their stated multiplicity adjustment. Municipal totals cannot reveal which individuals switched candidates, entered the electorate, or stopped voting. We therefore cannot establish persuasion rather than mobilization as the source of the mayor's gain.[20]

Complementary Services and Political Relationships. Offices that register families also provide assistance that families may value in its own right. The PAIF comparisons cannot separate appreciation of those services from credit for access to the federal transfer. Differences in who becomes enrolled could also explain why electoral gains vary across municipalities, even if contact has no independent effect. The sensitivity to municipal composition makes this possibility relevant. Political relationships may also change as families become less economically vulnerable: Frey (2019) interprets a negative local-incumbency response to transfer expansion through reduced opportunities for clientelistic exchange. Our positive estimate concerns a different source of expansion and population, and allows for offsetting changes in such relationships. We do not observe exchanges of benefits for votes.

4.6 Institutional Interpretation

The institutional link is between federal authorization and local access. The federal government finances the benefit, but municipal staff can help eligible families become visible to the program by identifying them, explaining its requirements, and assisting with registration. The increase in first registrations among families recorded within the income eligibility range is consistent with this role. It shows that expansion reaches be-

[20]The distinction also arises in the Progresa literature: De La O (2013) advances a mobilization interpretation, whereas Imai et al. (2020) show that corrected geographic matches and electoral denominators undermine the original electoral findings. We retain both studies in Appendix A; their disagreement cautions against inferring individual behavior from aggregate electoral contrasts.

yond the registry's existing stock. It does not establish that all additional beneficiaries were newly registered: families already in the registry could also obtain benefits when quotas expanded.

This local role offers a plausible connection between household gains and electoral support. A family may value both the transfer and the assistance that makes access possible. Where an office or outreach worker provides that assistance, the municipality's contribution may be visible even though the payment comes from the federal government. The CRAS and outreach comparisons motivate this interpretation, while the composition sensitivity and the less consistent evidence on contact practices and operating duration limit what we can attribute to those institutions. The comparisons support studying local assistance as a possible mechanism; they do not isolate the political return to a particular encounter or service.

The household evidence discussed in Section 3.6 explains why facilitating access could have value for families as well as for officeholders. Electoral rewards and improved access to social protection can coexist. Our results are consistent with that alignment, but do not show how much municipal effort was motivated by the prospect of reelection. Nor do they identify the share of transfers or votes that would disappear if new registration had not occurred. The conditional registration scenarios therefore describe the consequences of assumed links between registration, receipt, and votes; they do not supply the missing causal decomposition. The substantive finding is that federal expansion has a local political return in a setting where municipal assistance can help families obtain a valuable benefit.

5 Conclusion

This paper studies the electoral consequences of expanding a federal cash-transfer program whose implementation depends on municipal governments. We use changes in federal enrollment quotas to examine how local expansion of *Bolsa Família* affects support for mayors seeking reelection. Greater authorization brought more families into the program and increased incumbent vote share. Our preferred estimate implies an electoral gain of 4.69 percentage points for a one-standard-deviation increase in enrollment induced by the quota reform.

The findings highlight the political importance of the institutions through which families obtain public benefits. Federal authorities determine eligibility and make payments, while municipal staff identify eligible households and help them register. The

evidence on local welfare offices is consistent with contact between families and municipal staff contributing to electoral support. Families may credit officials who help them obtain assistance, value the services provided alongside the transfer, or respond to the income gains themselves. Understanding how families perceive these different contributions would help explain who receives political credit for a shared policy responsibility.

The broader implication is that the electoral consequences of redistribution depend on how programs reach their beneficiaries. Centralized rules and payments leave an important role for local governments in making benefits accessible. Our findings show that municipal incumbents can gain electoral support through the expansion of a federally financed program, placing local implementation at the center of the political economy of social protection.

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Figures and Tables

Table 1: *Bolsa Família* Eligibility Thresholds and Benefit Values, 2004–2018

	2004 Jan.	2006 Apr.	2007 Aug.	2008 Mar.	2008 July	2009 Apr.	2009 July	2009 Sept.	2011 Apr.	2011 Sept.	2011 Nov.	2011 Dec.	2012 June	2012 Dec.	2013 Mar.	2014 May	2014 June	2016 June	2016 July	2018 July
Panel A: Monthly income eligibility thresholds (BRL per capita)																				
Extreme-poverty line	50	60	60	60	60	69	70	70	70	70	70	70	70	70	70	77	77	85	85	89
Poverty line	100	120	120	120	120	137	140	140	140	140	140	140	140	140	140	154	154	170	170	178
Panel B: Monthly benefit values (current BRL)																				
Basic benefit (per family)	50	50	58	58	62	62	62	68	70	70	70	70	70	70	70	70	77	77	85	89
Variable benefit (per child aged 0–15)	15	15	18	18	20	20	20	22	32	32	32	32	32	32	32	32	35	35	39	41
Youth benefit (per member aged 16–17)	–	–	–	30	30	30	30	33	38	38	38	38	38	38	38	38	42	42	46	48
Pregnancy benefit (per pregnancy)	–	–	–	–	–	–	–	–	–	–	–	32	32	32	32	32	35	35	39	41
Nursing benefit (per eligible infant)	–	–	–	–	–	–	–	–	–	–	32	32	32	32	32	32	35	35	39	41
Extreme-poverty supplement (per family)	–	–	–	–	–	–	–	–	–	–	–	–	G	G	G	G	G	G	G	G
Legacy-program compensation	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T
Panel C: Family benefit limits and payment duration																				
Variable benefits (shared family limit)	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5	5	5	5	5	5
Youth benefits (family limit)	–	–	–	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Pregnancy benefit (monthly installments)	–	–	–	–	–	–	–	–	–	–	–	9	9	9	9	9	9	9	9	9
Nursing benefit (monthly installments)	–	–	–	–	–	–	–	–	–	–	6	6	6	6	6	6	6	6	6	6
Supplement (family limit)	–	–	–	–	–	–	–	–	–	–	–	–	1	1	1	1	1	1	1	1
Supplement: required child age	–	–	–	–	–	–	–	–	–	–	–	–	0–6	0–15	Any	Any	Any	Any	Any	Any

Notes: Columns report effective months: legal eligibility dates for income thresholds and first payment months for benefit changes, rather than announcement dates. Values are nominal; unchanged rules carry forward through December 2018. A dash indicates a component not yet implemented. Basic benefits apply to extremely poor families. Child, pregnancy and nursing benefits share the variable-benefit family limit; the latter two are not additional slots. Pregnancy/nursing benefits pay nine/six installments per eligible episode. G is a family-specific supplement: family size multiplied by (the applicable target minus post-transfer income per capita), rounded up to the next BRL 2 multiple. Targets are BRL 70.01, 77.01, 85.01 and 89.01. Its target changes with benefit-payment dates. Any removes the child-composition requirement, not the income test. T preserves qualifying predecessor-program payments above the program ceiling; it is family-specific, rounded up to the next BRL. Subnational supplements are outside this federal schedule.

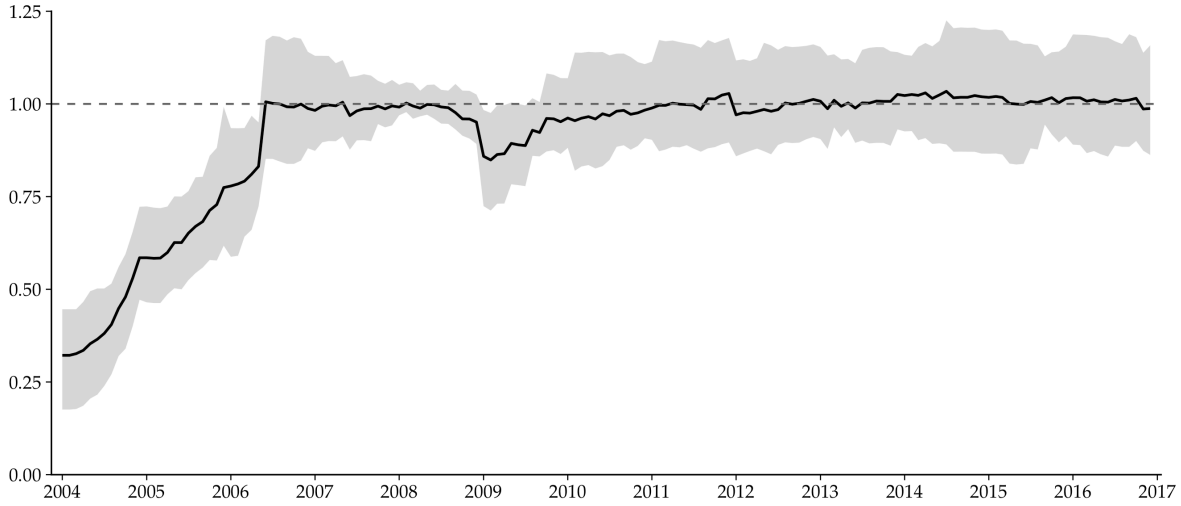
Sources: Jan. 2004 baseline: Law 10,836; legacy compensation: Decree 5,209 (art. 19). Apr. 2006 thresholds: Decree 5,749; Aug. 2007 amounts: Decree 6,157. Mar. 2008 youth benefit: Provisional Measure 411 and MDS Informe 158; July 2008 amounts: Decree 6,491. Apr. 2009 thresholds: Decree 6,824; July thresholds/Sept. amounts: Decree 6,917. Apr. 2011 amounts: Decree 7,447; Sept. family limit: Decree 7,494 and MDS Informe 285; Nov./Dec. nursing/pregnancy implementation: MDS Informe 293. June/Dec. 2012 supplement: Decree 7,758/Decree 7,852; Mar. 2013 extension: Law 12,817 (art. 2). May thresholds/June amounts, 2014: Decree 8,232; June thresholds/July amounts, 2016: Decree 8,794; July 2018 thresholds and amounts: Decree 9,396.

Table 2: Descriptive Statistics

Panel A: All Municipalities							**Panel B: Regression Sample**						
	N	Min	Mean	Median	Max	SD		N	Min	Mean	Median	Max	SD
Population (2000)	4,976	795.00	31,515.59	10,703.50	10,277,518.00	190,735.22	Population (2000)	2,078	795.00	27,991.14	10,556.50	5,773,130.00	146,177.39
Urban Population (2000) (%)	4,976	0.00	59.29	59.92	100.00	23.32	Urban Population (2000) (%)	2,078	0.00	58.80	59.35	100.00	23.32
Poor Population (2000) (%)	4,976	5.17	64.02	66.23	99.00	20.76	Poor Population (2000) (%)	2,078	7.66	64.09	66.80	99.00	20.99
Income per capita (2000) (R$)	4,976	62.65	338.27	306.00	1,759.76	193.29	Income per capita (2000) (R$)	2,078	69.20	337.40	300.13	1,759.76	195.41
Gini (2000) (%)	4,976	30.00	54.78	55.00	82.00	6.80	Gini (2000) (%)	2,078	30.00	54.84	55.00	80.00	6.73
Formal Workers (2000) (%)	4,976	1.92	36.15	34.44	86.38	18.16	Formal Workers (2000) (%)	2,078	2.89	35.80	33.98	86.38	18.31
Permitted *Bolsa Família* Families (2008)	4,976	24.00	2,078.06	917.50	285,060.00	6,980.68	Permitted *Bolsa Família* Families (2008)	2,078	24.00	1,898.41	906.00	164,913.00	5,077.26
Permitted *Bolsa Família* Families (2008) (%)	4,976	0.77	8.81	8.29	67.43	4.66	Permitted *Bolsa Família* Families (2008) (%)	2,078	0.77	8.89	8.39	33.00	4.71
Permitted *Bolsa Família* Families (2008) per 100 Families	4,976	2.89	34.32	30.33	113.31	18.66	Permitted *Bolsa Família* Families (2008) per 100 Families	2,078	2.89	34.60	31.12	110.30	18.94
Permitted *Bolsa Família* Families (2008) per 10,000 Residents	4,976	77.08	880.65	829.13	6,742.82	466.41	Permitted *Bolsa Família* Families (2008) per 10,000 Residents	2,078	77.08	888.88	839.23	3,299.89	470.55
Δ Permitted *Bolsa Família* Families per 100 Families	4,976	-36.28	5.59	3.48	103.48	11.72	Δ Permitted *Bolsa Família* Families per 100 Families	2,078	-36.28	5.71	3.57	86.01	11.74
Δ Permitted *Bolsa Família* Families per 10,000 Residents	4,976	-2,841.24	116.55	91.84	2,865.11	274.55	Δ Permitted *Bolsa Família* Families per 10,000 Residents	2,078	-1,000.62	120.42	95.75	2,865.11	277.26
Realized *Bolsa Família* Families (2008)	4,976	2.42	2,045.08	914.79	204,410.17	5,990.93	Realized *Bolsa Família* Families (2008)	2,078	5.50	1,871.27	908.38	133,322.75	4,543.18
Realized *Bolsa Família* Families (2008) (%)	4,976	0.08	8.76	8.13	61.49	4.91	Realized *Bolsa Família* Families (2008) (%)	2,078	0.26	8.80	8.31	29.64	4.90
Realized *Bolsa Família* Families (2008) per 100 Families	4,976	0.28	34.49	30.03	105.92	20.21	Realized *Bolsa Família* Families (2008) per 100 Families	2,078	0.93	34.58	30.84	89.54	20.26
Realized *Bolsa Família* Families (2008) per 10,000 Residents	4,976	8.48	876.40	813.07	6,149.19	491.21	Realized *Bolsa Família* Families (2008) per 10,000 Residents	2,078	25.81	879.87	830.99	2,963.95	490.46
Δ Realized *Bolsa Família* Families per 100 Families	4,976	-13.85	7.78	5.26	93.70	10.06	Δ Realized *Bolsa Família* Families per 100 Families	2,078	-12.71	8.63	6.09	93.70	10.28
Δ Realized *Bolsa Família* Families per 10,000 Residents	4,976	-1,234.13	180.99	139.04	2,979.12	231.70	Δ Realized *Bolsa Família* Families per 10,000 Residents	2,078	-398.59	203.50	163.15	2,979.12	238.38
Candidates (2008)	4,976	2.00	2.73	2.00	12.00	1.02	Candidates (2008)	2,078	2.00	2.80	3.00	12.00	1.07
Candidates (2012)	4,976	2.00	2.77	2.00	12.00	1.08	Candidates (2012)	2,078	2.00	2.70	2.00	9.00	1.00
Trying Reelection (2012) (%)	4,976	0.00	43.75	0.00	100.00	49.61	Trying Reelection (2012) (%)	2,078	100.00	100.00	100.00	100.00	0.00
Winner's Vote Share (2008) (%)	4,976	24.06	54.97	53.73	99.98	10.46	Winner's Vote Share (2008) (%)	2,078	24.06	52.92	52.68	99.16	9.43
Incumbent Vote Share (2012) (%)	2,177	0.00	48.28	49.24	100.00	16.41	Incumbent Vote Share (2012) (%)	2,078	0.18	49.99	49.65	100.00	14.03
Δ Incumbent Vote Share (pp)	2,177	-79.46	-4.60	-3.60	55.47	17.29	Δ Incumbent Vote Share (pp)	2,078	-71.82	-2.94	-3.03	55.47	15.13
Female Mayor (2008) (%)	4,970	0.00	9.22	0.00	100.00	28.93	Female Mayor (2008) (%)	2,078	0.00	10.15	0.00	100.00	30.21
High School or More (2008) (%)	4,970	0.00	78.85	100.00	100.00	40.84	High School or More (2008) (%)	2,078	0.00	81.14	100.00	100.00	39.13
College Degree (2008) (%)	4,970	0.00	45.11	0.00	100.00	49.77	College Degree (2008) (%)	2,078	0.00	46.39	0.00	100.00	49.88
Mayor Age (2008)	4,970	21.00	48.16	48.00	90.00	9.52	Mayor Age (2008)	2,078	21.00	46.51	46.00	81.00	9.08
Married (2008) (%)	4,970	0.00	78.13	100.00	100.00	41.34	Married (2008) (%)	2,078	0.00	77.05	100.00	100.00	42.06
Public-Sector Occupation (2008) (%)	4,970	0.00	34.21	0.00	100.00	47.44	Public-Sector Occupation (2008) (%)	2,078	0.00	15.74	0.00	100.00	36.42
Agricultural Occupation (2008) (%)	4,970	0.00	12.03	0.00	100.00	32.54	Agricultural Occupation (2008) (%)	2,078	0.00	14.49	0.00	100.00	35.20
Business Occupation (2008) (%)	4,970	0.00	20.24	0.00	100.00	40.18	Business Occupation (2008) (%)	2,078	0.00	26.32	0.00	100.00	44.05

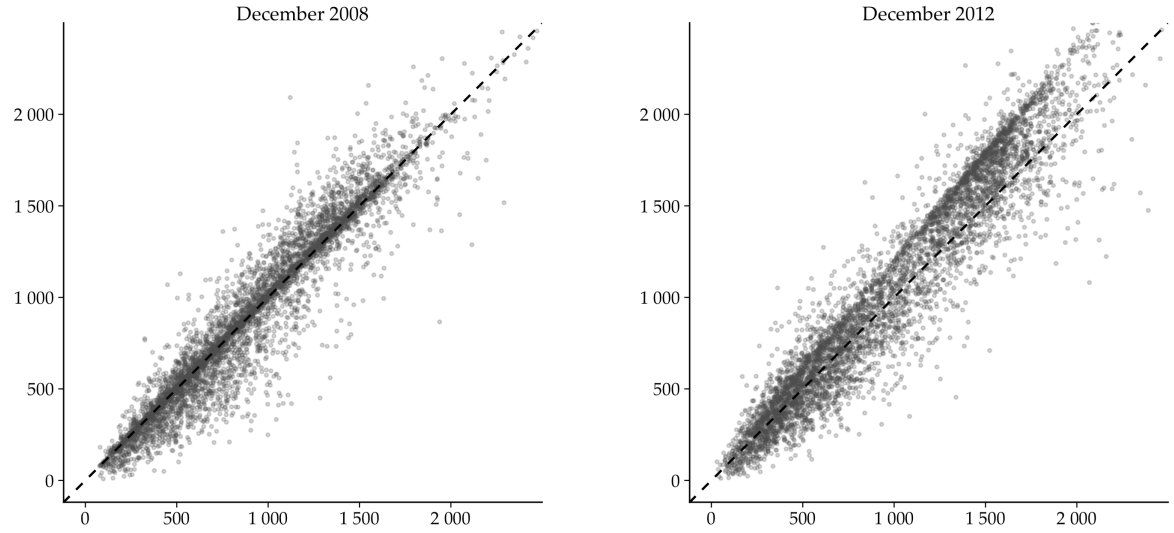
Notes: Each observation is one municipality. Panel A contains municipalities with a balanced 2008–2012 election panel that satisfy the common election-quality restrictions; *N* is the number with nonmissing values in each row. Panel B is the exact locked regression sample used in the headline estimates (2,078 municipalities). Permitted families in 2008 are the pre-reform authorized quota (EST 2006); realized families are 12-month averages. Percentage and per-10,000 measures use fixed 2006 population, while per-100-family measures use Census-2000 families. Every Δ is the corresponding 2012 value minus its 2008 value. Candidate counts describe first-round mayoral candidates. Trying reelection indicates that the sitting mayor contested the 2012 election. Vote share tracks the 2008 winner: the 2008 value is that winner's baseline share, and the 2012 value and its change are observed only when the same mayor reran. Mayor characteristics describe the 2008 winner and match the headline controls; age squared is omitted because it is a deterministic transformation of age.

Figure 1: Realized Enrollment as a Share of Authorized Quota



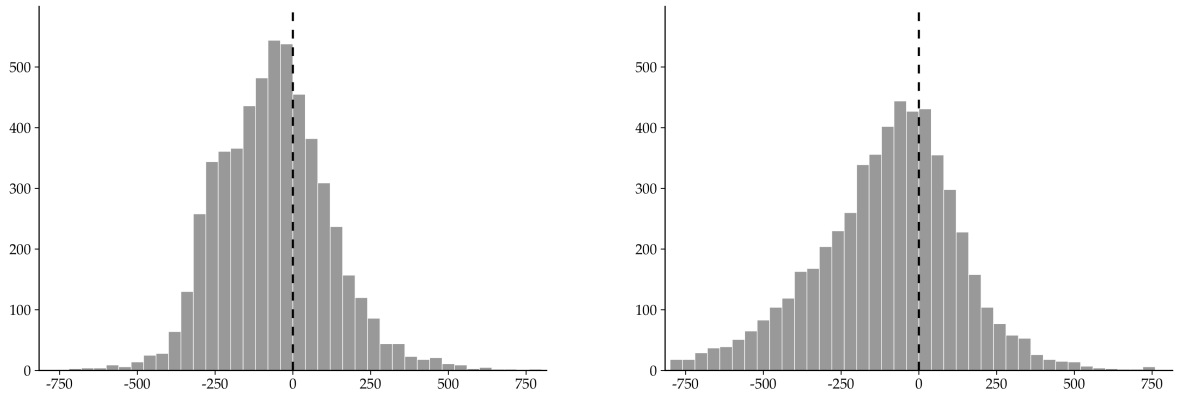
Note. For each municipality-month, we compute the ratio of realized *Bolsa Família* beneficiaries to the authorized quota. The solid line plots the quota-weighted cross-municipal mean of these municipality-level ratios. The shaded band spans the weighted 25th to 75th percentile. The dashed horizontal line marks full take-up (ratio = 1). Before 2007, municipalities operated well below their authorized ceilings, and the rapid convergence toward 1 reflects the administrative build-up of the program in its first years. After this initial convergence, realized enrollment generally tracked authorized capacity closely. The 2009 reform temporarily broke that alignment: expanded quotas opened a gap between authorized slots and realized beneficiaries, creating the local implementation margin through which municipalities had to identify, register, and enroll additional eligible families.

Figure 2: Authorized Quota vs. Realized Enrollment



Note. Each dot is a municipality. The horizontal axis shows the federally authorized quota and the vertical axis shows the 12-month average stock of families enrolled in *Bolsa Família*, both per 10,000 residents using fixed 2006 population. The dashed line marks the 45-degree locus of full take-up (realized = authorized). The panels show December 2008, before the mayoral term, and December 2012, at its endpoint. Both axes share a cap at the 99th percentile of the displayed cross-municipal distributions (approximately 2,400 beneficiary families per 10,000 residents); observations above this threshold are excluded from the frame but not from any analysis.

Figure 3: Distribution of Slack Between Authorized Quota and Realized Enrollment



Note. Each panel shows slack in December 2012 per 10,000 residents using fixed 2006 population. Slack is the relevant authorized quota minus the 12-month average stock of beneficiary families in 2012. The left panel uses the official 2012 quota; the right panel uses the counterfactual 2009 quota implied by the previous allocation rule. Positive values indicate authorized room not translated into realized enrollment, while negative values indicate realized enrollment above the relevant quota. The dashed vertical line marks zero slack. Both panels retain the common display window from -800 to 800 beneficiary families per 10,000 residents to limit the influence of extreme observations.

Table 3: First-Stage Estimates

	ΔBeneficiaries$_m$		
	(1)	(2)	(3)
ΔQuota$_m$ (π_1)	0.317	0.359	0.301
	(0.063)	(0.062)	(0.052)
Kleibergen–Paap F	25.43	33.63	33.30
Incumbent Party	✓	✓	✓
Bolsa Família Intensity	✓	✓	✓
Municipality Characteristics		✓	✓
Mayor Characteristics		✓	✓
Benefit Value			✓
FPM Transfers			✓
Number of Mayors	2,078	2,078	2,078
Observations	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes incumbent-party and baseline program-intensity controls; Column (2) adds municipality and mayor characteristics; Column (3) adds 2008 benefit value and FPM transfers, each interacted with post. Standard errors are in parentheses.

Table 4: Reduced-Form Estimates

	Δ Incumbent's Vote Share			Δ Turnout		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔQuota_m (ρ_1)	1.929	1.299	1.414	-0.372	-0.352	-0.354
	(0.604)	(0.461)	(0.482)	(0.192)	(0.181)	(0.180)
Incumbent Party	✓	✓	✓	✓	✓	✓
Bolsa Família Intensity	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓	✓		✓	✓
Mayor Characteristics		✓	✓		✓	✓
Benefit Value			✓			✓
FPM Transfers			✓			✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes incumbent-party and baseline program-intensity controls; Column (2) adds municipality and mayor characteristics; Column (3) adds 2008 benefit value and FPM transfers, each interacted with post. Standard errors are in parentheses.

Table 5: Two-Stage Least Squares Estimates

	Δ Incumbent's Vote Share			Δ Turnout		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	6.082	3.619	4.690	-1.173	-0.981	-1.173
	(2.322)	(1.569)	(1.991)	(0.698)	(0.532)	(0.639)
AR 95% CI	[2.019, 12.834]	[0.876, 7.978]	[1.242, 10.282]	[-3.168, 0.069]	[-2.302, 0.060]	[-2.792, 0.053]
Incumbent Party	✓	✓	✓	✓	✓	✓
Bolsa Família Intensity	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓	✓		✓	✓
Mayor Characteristics		✓	✓		✓	✓
Benefit Value			✓			✓
FPM Transfers			✓			✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes incumbent-party and baseline program-intensity controls; Column (2) adds municipality and mayor characteristics; Column (3) adds 2008 benefit value and FPM transfers, each interacted with post. Standard errors are in parentheses. The Kleibergen–Paap statistic is the squared state-clustered first-stage t-ratio. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness.

Table 6: Heterogeneity by Municipality Size

	Δ Incumbent's Vote Share			Δ Turnout		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Below Median						
ΔQuota_m (ρ_1)	-0.147	-0.057	0.033	-0.167	-0.172	-0.223
	(0.318)	(0.289)	(0.292)	(0.117)	(0.103)	(0.109)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-0.303	-0.114	0.073	-0.345	-0.347	-0.493
	(0.675)	(0.585)	(0.643)	(0.260)	(0.216)	(0.265)
AR 95% CI	[-1.979, 0.961]	[-1.473, 1.040]	[-1.435, 1.369]	[-0.985, 0.145]	[-0.851, 0.078]	[-1.164, 0.003]
Panel B: Below the 75th Percentile						
ΔQuota_m (ρ_1)	-0.078	-0.138	0.042	-0.290	-0.230	-0.291
	(0.315)	(0.329)	(0.291)	(0.198)	(0.159)	(0.156)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-0.200	-0.312	0.109	-0.740	-0.519	-0.755
	(0.800)	(0.740)	(0.756)	(0.571)	(0.384)	(0.448)
AR 95% CI	[-1.844, 1.662]	[-1.881, 1.275]	[-1.468, 1.790]	[-2.288, 0.267]	[-1.442, 0.208]	[-1.879, 0.075]
Panel C: Above the 75th Percentile						
ΔQuota_m (ρ_1)	3.873	2.584	2.908	-0.412	-0.487	-0.430
	(1.113)	(0.883)	(0.928)	(0.203)	(0.246)	(0.278)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	12.596	7.731	11.270	-1.341	-1.458	-1.666
	(4.061)	(3.173)	(4.840)	(0.775)	(0.772)	(1.158)
AR 95% CI	[5.274, 24.648]	[2.195, 17.645]	[3.368, 27.324]	[-3.703, 0.018]	[-3.592, 0.065]	[-4.931, 0.558]
Panel D: Above the 90th Percentile						
ΔQuota_m (ρ_1)	6.790	5.301	5.424	-0.346	-0.537	-0.448
	(1.425)	(0.947)	(1.048)	(0.204)	(0.312)	(0.345)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	20.717	14.167	18.257	-1.056	-1.434	-1.509
	(7.342)	(4.990)	(7.058)	(0.824)	(0.989)	(1.383)
AR 95% CI	[10.647, 71.349]	[7.342, 42.878]	[8.677, 60.479]	[-6.167, 0.200]	[-5.886, 0.294]	[-8.196, 0.813]
Incumbent Party	✓	✓	✓	✓	✓	✓
Bolsa Família Intensity	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓	✓		✓	✓
Mayor Characteristics		✓	✓		✓	✓
Benefit Value			✓			✓
FPM Transfers			✓			✓

Notes: Panels use the 2006-population median, 75th percentile, and 90th percentile calculated in the headline sample. The below-median and below-75th panels are nested; the above-90th-percentile panel is a subset of the above-75th panel. Models are estimated separately within each displayed sample. Columns follow the headline control sequence; all other specification choices match the main tables. No mean column is reported.

Table 7: Heterogeneity by Subgroup: Frey Sample and Baseline *Bolsa Família* Intensity

	Frey 2019			Baseline *Bolsa Família* Intensity		
	Outside Sample	Inside Sample	Δ	Below Median	Above Median	Δ
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.213	0.366	-0.153	0.255	0.370	-0.115
	(0.068)	(0.055)	(0.064)	(0.044)	(0.055)	(0.057)
Conditional First-Stage F	9.75	44.76		33.42	45.09	
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	3.017	-0.045	3.062	2.603	-0.150	2.753
	(0.962)	(0.270)	(0.994)	(1.056)	(0.222)	(1.162)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	14.198	-0.122	14.320	10.192	-0.405	10.598
	(7.255)	(0.736)	(7.213)	(4.301)	(0.612)	(4.599)
Conditional AR 95% CI	[3.867, 52.461]	[-1.706, 1.483]		[1.723, 20.805]	[-1.841, 0.815]	
Incumbent Party	✓	✓		✓	✓	
Bolsa Família Intensity	✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓	
Benefit Value	✓	✓		✓	✓	
FPM Transfers	✓	✓		✓	✓	
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: Each three-column block uses the full headline sample. The Frey sample comprises municipalities with 2000 population below 30,000 and 2010 HDI below 0.7. Each block is one fully interacted dual-IV regression. The instrument, endogenous variable, and complete control set are allowed to vary across the complementary subgroup indicators; Δ is the first coefficient minus the second. The saturated system reproduces separate subgroup coefficients while retaining one joint model for the displayed comparison. All models use 2006-population weights and state-clustered standard errors. Contrast columns are calculated from the joint model and therefore do not receive control check marks. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). Group AR sets are plug-in conditional diagnostics that condition on the other group coefficient, not exact joint weak-IV regions.

Table 8: Heterogeneity by Subgroup: Northeast States and PT Incumbent

	Northeast States			PT Incumbent		
	Other Regions	Northeast	Δ	Other Parties	PT	Δ
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.390	0.495	-0.105	0.280	0.365	-0.085
	(0.075)	(0.031)	(0.082)	(0.036)	(0.125)	(0.109)
Conditional First-Stage F	26.85	247.95		70.40	9.58	
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	1.059	0.502	0.556	1.814	-0.501	2.315
	(0.871)	(0.409)	(0.962)	(0.630)	(0.982)	(1.447)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	2.712	1.015	1.697	6.490	-1.374	7.864
	(2.396)	(0.801)	(2.526)	(2.716)	(2.678)	(4.838)
Conditional AR 95% CI	[-1.789, 9.135]	[-0.727, 2.604]		[1.666, 13.098]	[-10.589, 13.499]	
Incumbent Party	✓	✓		✓	✓	
Bolsa Família Intensity	✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓	
Benefit Value	✓	✓		✓	✓	
FPM Transfers	✓	✓		✓	✓	
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: Each three-column block uses the full headline sample. Each block is one fully interacted dual-IV regression. The instrument, endogenous variable, and complete control set are allowed to vary across the complementary subgroup indicators; Δ is the first coefficient minus the second. The saturated system reproduces separate subgroup coefficients while retaining one joint model for the displayed comparison. All models use 2006-population weights and state-clustered standard errors. Contrast columns are calculated from the joint model and therefore do not receive control check marks. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). Group AR sets are plug-in conditional diagnostics that condition on the other group coefficient, not exact joint weak-IV regions.

Table 9: Robustness Checks: Vote Share

	Δ Incumbent's Vote Share						
		Alt. Outcomes		Alt. Shock		Alt. Weights	Alt. Scale
	Main	Winsorized	Registered Votes	Winsorized	Limited to 2009	IPW	Per Families
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: First Stage							
ΔQuota_m (π_1)	0.301	0.301	0.301	0.277	0.236	0.302	0.281
	(0.052)	(0.052)	(0.052)	(0.043)	(0.034)	(0.049)	(0.048)
Kleibergen–Paap F	33.30	33.30	33.30	41.26	48.07	37.68	34.88
Panel B: Reduced Form							
ΔQuota_m (ρ_1)	1.414	1.277	1.145	1.523	0.584	1.250	1.295
	(0.482)	(0.429)	(0.407)	(0.478)	(0.621)	(0.452)	(0.507)
Panel C: Two-Stage Least Squares							
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	4.690	4.234	3.797	5.500	2.474	4.144	4.604
	(1.991)	(1.773)	(1.654)	(2.082)	(2.570)	(1.758)	(2.236)
AR 95% CI	[1.242, 10.282]	[1.164, 9.209]	[0.914, 8.408]	[1.782, 11.012]	[-3.245, 7.846]	[0.978, 8.809]	[0.772, 10.906]
Incumbent Party	✓	✓	✓	✓	✓	✓	✓
Bolsa Família Intensity	✓	✓	✓	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓	✓	✓	✓
Benefit Value	✓	✓	✓	✓	✓	✓	✓
FPM Transfers	✓	✓	✓	✓	✓	✓	✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156	4,156

Notes: Column (1) is the preferred valid-vote specification. Columns (2)–(3) winsorize the outcome at the 1st and 99th percentiles or divide incumbent votes by registered voters. Column (4) winsorizes the quota shock. Column (5) replaces the 2012 authorization with the corresponding 2009 authorization and retains the 2008–2012 beneficiary change. Column (6) combines inverse-selection and population weights. Column (7) expresses the instrument and treatment per 100 Census-2000 families. Every column includes the complete control set and state-clustered standard errors. AR sets use cluster-robust quadratic inversion.

Table 10: Continuous-Quota and Discrete-Beneficiary Specifications

	Δ Incumbent's Vote Share			Δ Turnout		
	(1)	(2)	(3)	(4)	(5)	(6)
Reduced Form						
ΔQuota_m (ρ_1)	1.929	1.299	1.414	-0.372	-0.352	-0.354
	(0.604)	(0.461)	(0.482)	(0.192)	(0.181)	(0.180)
Panel A: Continuous						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	6.082	3.619	4.690	-1.173	-0.981	-1.173
	(2.322)	(1.569)	(1.991)	(0.698)	(0.532)	(0.639)
AR 95% CI	[2.019, 12.834]	[0.876, 7.978]	[1.242, 10.282]	[-3.168, 0.069]	[-2.302, 0.060]	[-2.792, 0.053]
Panel B: Above Median						
$\mathbf{1}\{\Delta\widehat{\text{Beneficiaries}}_m \geq c_D\}$ (β_1)	30.129	15.579	21.951	-5.812	-4.223	-5.488
	(14.379)	(7.838)	(11.340)	(3.641)	(1.953)	(2.558)
AR 95% CI	[9.445, 123.527]	[3.430, 44.533]	[5.090, 67.370]	[-25.814, 0.379]	[-9.013, 0.335]	[-12.189, 0.328]
Panel C: Top 25 Percent						
$\mathbf{1}\{\Delta\widehat{\text{Beneficiaries}}_m \geq c_D\}$ (β_1)	24.045	13.966	17.647	-4.638	-3.786	-4.412
	(8.792)	(5.342)	(6.303)	(2.742)	(2.129)	(2.380)
AR 95% CI	[8.279, 49.109]	[3.826, 28.051]	[5.440, 33.661]	[-12.464, 0.273]	[-9.444, 0.225]	[-10.448, 0.204]
Panel D: Above the 90th Percentile						
$\mathbf{1}\{\Delta\widehat{\text{Beneficiaries}}_m \geq c_D\}$ (β_1)	29.649	18.877	22.446	-5.719	-5.117	-5.612
	(12.181)	(7.744)	(8.693)	(3.565)	(3.176)	(3.431)
AR 95% CI	[9.474, 69.956]	[4.972, 41.702]	[6.668, 47.813]	[-17.252, 0.322]	[-15.024, 0.273]	[-16.290, 0.228]
Incumbent Party	✓	✓	✓	✓	✓	✓
Bolsa Família Intensity	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓	✓		✓	✓
Mayor Characteristics		✓	✓		✓	✓
Benefit Value			✓			✓
FPM Transfers			✓			✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: The standardized quota shock remains continuous in every panel. Panel A also retains the continuous beneficiary change. Panels B–D instead define the endogenous variable as an indicator that the municipality's 2008–2012 beneficiary change is at or above the sample median, 75th percentile, or 90th percentile, respectively. The reduced form is identical across panels by construction and is shown once above them. The 2SLS coefficient in Panels B–D is the effect of crossing the displayed beneficiary-expansion threshold. Each column follows the headline control sequence. Weights, clustering, and AR inversion otherwise match the main tables.

Table 11: Registration Expansion and Electoral Outcomes

	Incumbent Vote Share			Turnout		
	(1)	(2)	(3)	(4)	(5)	(6)
Reduced Form						
ΔQuota_m (ρ_1)	1.929	1.299	1.414	-0.372	-0.352	-0.354
	(0.604)	(0.461)	(0.482)	(0.192)	(0.181)	(0.180)
Panel A: All registered families						
$\Delta\widehat{\text{Registrations}}_m$ (β_1)	15.773	8.085	8.401	-3.043	-2.191	-2.101
	(7.453)	(3.933)	(4.039)	(2.037)	(1.297)	(1.224)
AR 95% confidence set	[4.819, 55.175]	[1.867, 22.852]	[2.076, 23.020]	[-13.075, 0.170]	[-6.495, 0.130]	[-5.972, 0.093]
Panel B: Income above R$140						
$\Delta\widehat{\text{Registrations}}_m$ (β_1)	-50.197	-87.908	36.512	9.683	23.829	-9.129
	(56.262)	(225.199)	(34.739)	(12.442)	(61.588)	(8.783)
AR 95% confidence set	$(-\infty, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$
Panel C: Income at or below R$140						
$\Delta\widehat{\text{Registrations}}_m$ (β_1)	11.680	6.562	7.922	-2.253	-1.779	-1.981
	(4.864)	(3.099)	(3.765)	(1.459)	(1.033)	(1.163)
AR 95% confidence set	[3.772, 29.293]	[1.512, 16.719]	[1.954, 20.788]	[-7.528, 0.123]	[-4.802, 0.104]	[-5.524, 0.086]
Panel D: Income at or below R$70						
$\Delta\widehat{\text{Registrations}}_m$ (β_1)	12.368	7.069	9.197	-2.386	-1.916	-2.299
	(5.082)	(3.265)	(4.224)	(1.538)	(1.113)	(1.364)
AR 95% confidence set	[4.011, 30.094]	[1.655, 17.378]	[2.338, 23.106]	[-7.776, 0.130]	[-5.117, 0.111]	[-6.450, 0.098]
Incumbent Party	✓	✓	✓	✓	✓	✓
Baseline *Bolsa Família* Intensity	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓	✓		✓	✓
Mayor Characteristics		✓	✓		✓	✓
Benefit Value			✓			✓
FPM Transfers			✓			✓
Municipalities	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns retain the locked 2008–2012 sample, quota-shock instrument, municipality and year fixed effects, 2006-population weights and state-clustered CRV1 standard errors. Controls are interacted with post. Registration expansion counts families first registered in 2009–2012 and still observed in the 2012 *Cadastro Único* snapshot. Income (monthly per capita) and municipality are measured in 2012. These are surviving entry cohorts, not historical net stocks or verified *Bolsa Família* recipients. The income definitions overlap: the at-or-below-R$140 group combines the two lower-income bands. Each endogenous registration measure uses its own fixed weighted standard deviation. Reduced forms are common; these IVs do not separate registration from subsequent transfers. Parentheses report state-clustered standard errors. Disconnected two-tail Anderson–Rubin sets use the conservative full-real-line envelope; exact components remain in the result files.

Table 12: Heterogeneity by CRAS Presence

	Δ Incumbent's Vote Share			Δ Turnout		
	Without CRAS	With CRAS	Δ	Without CRAS	With CRAS	Δ
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.325	0.301	0.023	0.325	0.301	0.023
	(0.040)	(0.060)	(0.056)	(0.040)	(0.060)	(0.056)
Conditional First-Stage F	66.76	25.05		66.76	25.05	
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	-0.398	1.841	-2.239	-0.138	-0.385	0.247
	(0.634)	(0.540)	(0.736)	(0.073)	(0.213)	(0.206)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-1.226	6.111	-7.338	-0.425	-1.277	0.852
	(1.935)	(2.394)	(2.518)	(0.220)	(0.765)	(0.733)
Conditional AR 95% CI	[-5.247, 2.991]	[2.123, 13.449]		[-0.897, 0.039]	[-3.332, 0.176]	
Incumbent Party	✓	✓		✓	✓	
Bolsa Família Intensity	✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓	
Benefit Value	✓	✓		✓	✓	
FPM Transfers	✓	✓		✓	✓	
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: Each outcome is estimated in one fully interacted dual-IV system allowing the standardized instrument, treatment, and complete control set to vary with 2008 *CRAS* presence and its complement; Δ is the first coefficient minus the second. The saturated system reproduces separate subgroup coefficients while retaining one joint model for the displayed comparison. Models use the headline sample, 2006-population weights, and state-clustered standard errors. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). AR sets are plug-in conditional diagnostics, not exact joint weak-IV regions.

Table 13: Descriptive Statistics by CRAS Presence

Panel A: Municipalities With CRAS	N	Min	Mean	Median	Max	SD
Population (2000)	1,394	1,262.00	36,787.28	13,389.00	5,773,130.00	177,650.70
Urban Population (2000) (%)	1,394	4.67	59.52	59.54	100.00	23.56
Poor Population (2000) (%)	1,394	7.66	67.53	74.39	99.00	20.71
Income per capita (2000) (R$)	1,394	69.20	315.16	243.12	1,235.29	197.96
Gini (2000) (%)	1,394	35.00	55.79	56.00	79.00	6.26
Formal Workers (2000) (%)	1,394	2.89	33.69	30.49	80.88	18.25
Permitted *Bolsa Família* Families (2008)	1,394	58.00	2,485.34	1,351.00	164,913.00	6,090.08
Permitted *Bolsa Família* Families (2008) (%)	1,394	0.82	9.56	9.48	30.25	4.70
Permitted *Bolsa Família* Families (2008) per 100 Families	1,394	3.36	38.18	39.38	96.46	19.16
Permitted *Bolsa Família* Families (2008) per 10,000 Residents	1,394	81.88	956.42	948.41	3,025.26	469.94
Δ Permitted *Bolsa Família* Families per 100 Families	1,394	-36.28	7.03	4.98	84.50	11.93
Δ Permitted *Bolsa Família* Families per 10,000 Residents	1,394	-1,000.62	149.99	126.26	2,686.38	267.25
Realized *Bolsa Família* Families (2008)	1,394	23.67	2,455.52	1,339.92	133,322.75	5,424.96
Realized *Bolsa Família* Families (2008) (%)	1,394	0.82	9.65	9.80	24.91	4.88
Realized *Bolsa Família* Families (2008) per 100 Families	1,394	2.85	38.82	40.54	89.54	20.44
Realized *Bolsa Família* Families (2008) per 10,000 Residents	1,394	82.11	964.74	979.67	2,491.38	488.26
Δ Realized *Bolsa Família* Families per 100 Families	1,394	-12.71	9.94	7.92	93.70	10.52
Δ Realized *Bolsa Família* Families per 10,000 Residents	1,394	-398.59	230.89	203.79	2,979.12	235.14
Candidates (2008)	1,394	2.00	2.90	3.00	12.00	1.16
Candidates (2012)	1,394	2.00	2.82	3.00	9.00	1.07
Trying Reelection (2012) (%)	1,394	100.00	100.00	100.00	100.00	0.00
Winner's Vote Share (2008) (%)	1,394	24.06	52.90	52.60	99.16	9.91
Incumbent Vote Share (2012) (%)	1,394	0.18	49.82	49.22	100.00	14.64
Δ Incumbent Vote Share (pp)	1,394	-71.82	-3.08	-3.25	54.95	15.83
Female Mayor (2008) (%)	1,394	0.00	11.33	0.00	100.00	31.71
High School or More (2008) (%)	1,394	0.00	83.07	100.00	100.00	37.51
College Degree (2008) (%)	1,394	0.00	48.78	0.00	100.00	50.00
Mayor Age (2008)	1,394	21.00	46.57	46.00	78.00	9.27
Married (2008) (%)	1,394	0.00	75.75	100.00	100.00	42.87
Public-Sector Occupation (2008) (%)	1,394	0.00	15.93	0.00	100.00	36.60
Agricultural Occupation (2008) (%)	1,394	0.00	13.70	0.00	100.00	34.40
Business Occupation (2008) (%)	1,394	0.00	25.11	0.00	100.00	43.38

Panel B: Municipalities Without CRAS	N	Min	Mean	Median	Max	SD
Population (2000)	684	795.00	10,064.50	6,421.50	122,377.00	11,514.12
Urban Population (2000) (%)	684	0.00	57.33	59.09	100.00	22.77
Poor Population (2000) (%)	684	10.91	57.07	56.78	96.97	19.79
Income per capita (2000) (R$)	684	76.77	382.74	368.90	1,759.76	182.02
Gini (2000) (%)	684	30.00	52.92	53.00	80.00	7.22
Formal Workers (2000) (%)	684	4.74	40.09	39.79	86.38	17.69
Permitted *Bolsa Família* Families (2008)	684	24.00	702.24	457.00	6,344.00	782.52
Permitted *Bolsa Família* Families (2008) (%)	684	0.77	7.51	6.97	33.00	4.41
Permitted *Bolsa Família* Families (2008) per 100 Families	684	2.89	27.31	24.31	110.30	16.21
Permitted *Bolsa Família* Families (2008) per 10,000 Residents	684	77.08	751.21	696.50	3,299.89	441.17
Δ Permitted *Bolsa Família* Families per 100 Families	684	-24.84	3.02	0.96	86.01	10.87
Δ Permitted *Bolsa Família* Families per 10,000 Residents	684	-889.25	60.15	26.58	2,865.11	287.54
Realized *Bolsa Família* Families (2008)	684	5.50	680.56	425.79	6,362.42	792.05
Realized *Bolsa Família* Families (2008) (%)	684	0.26	7.07	6.32	29.64	4.48
Realized *Bolsa Família* Families (2008) per 100 Families	684	0.93	25.95	22.09	77.18	16.89
Realized *Bolsa Família* Families (2008) per 10,000 Residents	684	25.81	706.89	631.86	2,963.95	447.92
Δ Realized *Bolsa Família* Families per 100 Families	684	-8.84	5.94	3.43	53.99	9.22
Δ Realized *Bolsa Família* Families per 10,000 Residents	684	-301.46	147.68	95.65	2,440.54	235.35
Candidates (2008)	684	2.00	2.59	2.00	7.00	0.81
Candidates (2012)	684	2.00	2.44	2.00	8.00	0.78
Trying Reelection (2012) (%)	684	100.00	100.00	100.00	100.00	0.00
Winner's Vote Share (2008) (%)	684	28.05	52.97	52.98	98.18	8.37
Incumbent Vote Share (2012) (%)	684	5.05	50.33	50.61	100.00	12.70
Δ Incumbent Vote Share (pp)	684	-43.28	-2.64	-2.40	55.47	13.62
Female Mayor (2008) (%)	684	0.00	7.75	0.00	100.00	26.76
High School or More (2008) (%)	684	0.00	77.19	100.00	100.00	41.99
College Degree (2008) (%)	684	0.00	41.52	0.00	100.00	49.31
Mayor Age (2008)	684	23.00	46.38	46.00	81.00	8.68
Married (2008) (%)	684	0.00	79.68	100.00	100.00	40.27
Public-Sector Occupation (2008) (%)	684	0.00	15.35	0.00	100.00	36.07
Agricultural Occupation (2008) (%)	684	0.00	16.08	0.00	100.00	36.76
Business Occupation (2008) (%)	684	0.00	28.80	0.00	100.00	45.32

Notes: Both panels partition the locked headline sample by CRAS presence before the reform. Variables, scaling, and unweighted summary statistics match the main descriptive table; N counts nonmissing municipalities. The table includes baseline characteristics and the explicitly dated program and electoral outcomes shown in that table. Differences are descriptive, not estimates of the effect of CRAS.

Table 14: Heterogeneity by CRAS Intensity: Active Outreach

	Δ Incumbent's Vote Share			Δ Turnout		
	Without Active Outreach	With Active Outreach	Δ	Without Active Outreach	With Active Outreach	Δ
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.436	0.276	0.160	0.436	0.276	0.160
	(0.108)	(0.042)	(0.084)	(0.108)	(0.042)	(0.084)
Conditional First-Stage F	16.23	43.13		16.23	43.13	
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	0.757	2.004	-1.247	-0.234	-0.394	0.160
	(0.742)	(0.646)	(0.897)	(0.167)	(0.243)	(0.155)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	1.738	7.273	-5.535	-0.538	-1.430	0.892
	(1.841)	(2.572)	(2.702)	(0.437)	(0.894)	(0.647)
Conditional AR 95% CI	[-1.697, 7.407]	[2.401, 13.661]		[-1.964, 0.231]	[-3.487, 0.397]	
Incumbent Party	✓	✓		✓	✓	
Bolsa Família Intensity	✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓	
Benefit Value	✓	✓		✓	✓	
FPM Transfers	✓	✓		✓	✓	
Number of Mayors	1,394	1,394	1,394	1,394	1,394	1,394
Observations	2,788	2,788	2,788	2,788	2,788	2,788

Notes: Each outcome is estimated within 2008 *CRAS* municipalities in one fully interacted dual-IV system comparing units with and without active outreach; Δ is Without Active Outreach minus With Active Outreach. The instrument, treatment, and complete control set can vary across the two groups. Models use 2006-population weights and state-clustered standard errors. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). AR sets are plug-in conditional diagnostics, not exact joint weak-IV regions. Groups are predetermined scope conditions, not causal effects of adopting outreach.

Table 15: Heterogeneity by CRAS Intensity: *Proteção e Atendimento Integral à Família* (PAIF)

	Δ Incumbent's Vote Share			Δ Turnout		
	Without PAIF	With PAIF	Δ	Without PAIF	With PAIF	Δ
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.372	0.292	0.079	0.372	0.292	0.079
	(0.130)	(0.047)	(0.113)	(0.130)	(0.047)	(0.113)
Conditional First-Stage F	8.18	38.04		8.18	38.04	
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	0.653	2.251	-1.598	-0.442	-0.405	-0.037
	(0.570)	(0.791)	(1.004)	(0.279)	(0.204)	(0.162)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	1.758	7.697	-5.938	-1.189	-1.385	0.196
	(1.695)	(3.103)	(3.421)	(0.933)	(0.738)	(0.614)
Conditional AR 95% CI	[-1.501, 9.525]	[2.010, 15.783]		[-6.261, 0.324]	[-3.195, 0.050]	
Incumbent Party	✓	✓		✓	✓	
Bolsa Família Intensity	✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓	
Benefit Value	✓	✓		✓	✓	
FPM Transfers	✓	✓		✓	✓	
Number of Mayors	1,394	1,394	1,394	1,394	1,394	1,394
Observations	2,788	2,788	2,788	2,788	2,788	2,788

Notes: Each outcome is estimated within 2008 *CRAS* municipalities in one fully interacted dual-IV system comparing units with and without PAIF-tracked *Bolsa Família* families; Δ is Without PAIF minus With PAIF. The instrument, treatment, and complete control set can vary across the two groups. Models use 2006-population weights and state-clustered standard errors. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). AR sets are plug-in conditional diagnostics, not exact joint weak-IV regions. Groups are predetermined scope conditions, not causal effects of adopting PAIF services.

Table 16: Heterogeneity by CRAS Intensity: *Proteção e Atendimento Integral à Família* (PAIF) Volume

	Δ Incumbent's Vote Share			Δ Turnout		
	Below Median	Above Median	Δ	Below Median	Above Median	Δ
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.328	0.285	0.043	0.328	0.285	0.043
	(0.067)	(0.046)	(0.075)	(0.067)	(0.046)	(0.075)
Conditional First-Stage F	24.13	38.25		24.13	38.25	
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	-0.495	5.361	-5.856	-0.356	-0.384	0.028
	(0.958)	(1.407)	(2.026)	(0.350)	(0.137)	(0.343)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-1.509	18.778	-20.287	-1.085	-1.344	0.259
	(2.871)	(6.060)	(7.971)	(1.045)	(0.611)	(1.141)
Conditional AR 95% CI	[-7.706, 5.337]	[8.009, 35.047]		[-3.462, 1.278]	[-3.059, -0.306]	
Incumbent Party	✓	✓		✓	✓	
Bolsa Família Intensity	✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓	
Benefit Value	✓	✓		✓	✓	
FPM Transfers	✓	✓		✓	✓	
Number of Mayors	962	962	962	962	962	962
Observations	1,924	1,924	1,924	1,924	1,924	1,924

Notes: The sample contains 2008 *CRAS* municipalities reporting a positive number of *Bolsa Família* families followed through PAIF. Low and high volume are defined relative to the median positive family count in the headline sample; Δ is Below Median minus Above Median. The fully interacted dual-IV system allows the instrument, treatment, and complete control set to vary across the two groups. Models use the headline weights and state clustering. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016).

Table 17: Electoral Responses by CRAS Age Before the Reform

	Incumbent Vote Share			Turnout		
	No CRAS (1)	Opened by 2006 (2)	Opened 2007–2008 (3)	No CRAS (4)	Opened by 2006 (5)	Opened 2007–2008 (6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.325	0.269	0.408	0.325	0.269	0.408
	(0.040)	(0.043)	(0.097)	(0.040)	(0.043)	(0.097)
First-stage F	65.627	40.043	17.729	65.627	40.043	17.729
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	-0.398	2.446	1.002	-0.138	-0.373	-0.427
	(0.640)	(0.708)	(0.544)	(0.073)	(0.247)	(0.169)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-1.226	9.076	2.454	-0.425	-1.385	-1.047
	(1.951)	(2.991)	(1.657)	(0.222)	(0.908)	(0.566)
AR 95% confidence set	[-5.32, 3.08]	[3.55, 16.75]	[-0.25, 8.15]	[-0.91, 0.05]	[-3.42, 0.53]	[-3.07, -0.16]
Incumbent Party	✓	✓	✓	✓	✓	✓
Baseline *Bolsa Família* Intensity	✓	✓	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓	✓	✓
Benefit Value	✓	✓	✓	✓	✓	✓
FPM Transfers	✓	✓	✓	✓	✓	✓
Municipalities	684	831	563	684	831	563

Notes: Models use the locked two-election sample, full baseline controls, municipality and year fixed effects, and 2006-population weights. Parentheses report state-clustered standard errors. IV coefficients are percentage points per one fixed headline-sample weighted standard deviation of beneficiary-family expansion per 10,000 residents. Disconnected two-tail Anderson–Rubin 95% sets use their conservative full-real-line envelope; exact components remain in the sidecars. Subgroup comparisons do not identify a causal effect of CRAS. Opening years come from the baseline census. Missing dates are excluded. Separate subgroup fits reproduce the accepted fully interacted reduced-form slopes; their uncertainty uses each subgroup's own state count. First-stage treatment units are standardized as in the IV.

Table 18: Descriptive Quota Responses Across CRAS Adoption Cohorts

	First Stage		Reduced Form	
	Enrollment (1)	Registration (2)	Incumbent Votes (3)	Turnout (4)
Panel A: Opened by 2006				
ΔQuota_m (ρ_1)	48.874	30.121	2.446	-0.373
	(7.817)	(10.336)	(0.716)	(0.250)
95% confidence interval	[32.774, 64.973]	[8.833, 51.409]	[0.971, 3.921]	[-0.887, 0.141]
Panel B: Opened 2007–2008				
ΔQuota_m (ρ_1)	74.011	63.177	1.002	-0.427
	(17.674)	(18.510)	(0.547)	(0.170)
95% confidence interval	[37.611, 110.411]	[25.055, 101.298]	[-0.126, 2.129]	[-0.778, -0.077]
Panel C: Opened 2009–2010				
ΔQuota_m (ρ_1)	62.746	46.194	-0.612	-0.317
	(9.147)	(9.494)	(0.972)	(0.113)
95% confidence interval	[43.907, 81.584]	[26.640, 65.748]	[-2.614, 1.390]	[-0.549, -0.085]
Panel D: Opened 2011–September 2012				
ΔQuota_m (ρ_1)	78.589	96.626	-0.496	0.057
	(10.946)	(26.078)	(0.539)	(0.133)
95% confidence interval	[56.046, 101.132]	[42.918, 150.334]	[-1.607, 0.615]	[-0.216, 0.330]
Panel E: Not Observed in 2012				
ΔQuota_m (ρ_1)	42.752	39.928	0.779	-0.053
	(11.362)	(15.629)	(0.881)	(0.175)
95% confidence interval	[19.351, 66.153]	[7.740, 72.117]	[-1.036, 2.594]	[-0.414, 0.309]

Notes: Each cell reports the response to one fixed headline-sample weighted standard deviation of the quota shock, with state-clustered standard errors and pointwise confidence intervals. Enrollment and registrations are families per 10,000 residents; electoral outcomes are percentage points. Full baseline controls, municipality/year effects and population weights are retained. Later cohorts use dates reported by offices observed in 2012 and condition on post-reform choices. Not observed does not mean verified absence. These are descriptive subgroup responses, not causal effects of office openings. Contrasts and adjusted tests remain in the analysis sidecars.

Table 19: Where the Electoral Response Appears: Reduced Forms

	Pooled (1)	Without CRAS (2)	With CRAS (3)	Difference (4)
Panel A: Incumbent Votes				
ΔQuota_m (ρ_1)	1.145	-0.204	1.486	1.690
	(0.407)	(0.488)	(0.443)	(0.505)
Panel B: Other Candidates' Votes				
ΔQuota_m (ρ_1)	-1.205	0.331	-1.566	-1.897
	(0.391)	(0.530)	(0.452)	(0.671)
Panel C: Blank or Null Ballots				
ΔQuota_m (ρ_1)	-0.294	-0.265	-0.305	-0.040
	(0.194)	(0.189)	(0.252)	(0.302)
Panel D: Abstention				
ΔQuota_m (ρ_1)	0.354	0.138	0.385	0.247
	(0.180)	(0.073)	(0.213)	(0.206)
Panel E: Registered Voters				
ΔQuota_m (ρ_1)	120.393	110.898	124.979	14.080
	(34.967)	(22.257)	(40.831)	(38.417)
Incumbent Party	✓	✓	✓	✓
Baseline *Bolsa Família* Intensity	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓
Benefit Value	✓	✓	✓	✓
FPM Transfers	✓	✓	✓	✓

Notes: Models use the locked two-election sample, full baseline controls, municipality and year fixed effects, and 2006-population weights. Parentheses report state-clustered standard errors. IV coefficients are percentage points per one fixed headline-sample weighted standard deviation of beneficiary-family expansion per 10,000 residents. Disconnected two-tail Anderson–Rubin 95% sets use their conservative full-real-line envelope; exact components remain in the sidecars. Subgroup comparisons do not identify a causal effect of CRAS. The first four outcomes use registered voters as their denominator and sum to 100; their coefficients sum to zero. The last is registered voters per 10,000 residents. RF effects are per SD of quota exposure. Subgroup IVs use separate fits with subgroup-specific state counts. Aggregate accounting cannot identify individual voter transitions.

Table 20: Where the Electoral Response Appears: Two-Stage Least Squares

	Pooled (1)	Without CRAS (2)	With CRAS (3)
Panel A: Incumbent Votes			
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	3.797	-0.629	4.931
	(1.654)	(1.500)	(1.964)
AR 95% confidence set	[0.91, 8.41]	[-3.70, 2.77]	[1.67, 10.96]
Panel B: Other Candidates' Votes			
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-3.995	1.019	-5.197
	(1.634)	(1.642)	(1.983)
AR 95% confidence set	[-8.60, -1.18]	[-2.50, 4.57]	[-11.23, -1.87]
Panel C: Blank or Null Ballots			
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-0.975	-0.816	-1.011
	(0.672)	(0.632)	(0.886)
AR 95% confidence set	[-2.64, 0.35]	[-2.37, 0.38]	[-3.36, 0.69]
Panel D: Abstention			
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	1.173	0.425	1.277
	(0.639)	(0.222)	(0.763)
AR 95% confidence set	[-0.05, 2.79]	[-0.05, 0.91]	[-0.17, 3.32]
Panel E: Registered Voters			
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	399.329	341.608	414.847
	(92.847)	(72.923)	(105.773)
AR 95% confidence set	[194.37, 603.77]	[199.60, 514.99]	[174.36, 652.17]
Incumbent Party	✓	✓	✓
Baseline *Bolsa Família* Intensity	✓	✓	✓
Municipality Characteristics	✓	✓	✓
Mayor Characteristics	✓	✓	✓
Benefit Value	✓	✓	✓
FPM Transfers	✓	✓	✓
First-stage F	33.302	65.627	25.173
Municipalities	2078	684	1394

Notes: Models use the locked two-election sample, full baseline controls, municipality and year fixed effects, and 2006-population weights. Parentheses report state-clustered standard errors. IV coefficients are percentage points per one fixed headline-sample weighted standard deviation of beneficiary-family expansion per 10,000 residents. Disconnected two-tail Anderson–Rubin 95% sets use their conservative full-real-line envelope; exact components remain in the sidecars. Subgroup comparisons do not identify a causal effect of CRAS. The first four outcomes use registered voters as their denominator and sum to 100; their coefficients sum to zero. The last is registered voters per 10,000 residents. RF effects are per SD of quota exposure. Subgroup IVs use separate fits with subgroup-specific state counts. Aggregate accounting cannot identify individual voter transitions.

Table 21: Selection into Rerunning

	Incumbent Runs Again in 2012	
	(1)	(2)
ΔQuota$_m$	-2.183	0.045
	(1.156)	(1.017)
Incumbent Party	✓	✓
Bolsa Família Intensity	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors / Municipalities	5,492	5,492
Observations	5,492	5,492

Notes: The dependent variable equals 100 if the mayor elected in 2008 appears on the 2012 ballot and zero otherwise. The estimation universe is constructed before conditioning on rerunning. Every column is weighted by 2006 population and includes state fixed effects because municipality and year effects are not identified in this one-observation-per-mayor design. The quota shock is a rate per 10,000 residents using fixed 2006 population and is divided by the main-sample analysis-weighted SD. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. A † attached to a standard error denotes a wild-cluster $p < 0.05$, with resampling by state.

Table 22: Pre-Reform Vote-Share Balance

	Incumbent's Vote Share in 2008	
	(1)	(2)
ΔQuota$_m$	0.127	-0.070
	(0.254)	(0.228)
Incumbent Party	✓	✓
Bolsa Família Intensity	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors / Municipalities	2,078	2,078
Observations	2,078	2,078

Notes: The dependent variable is the vote share of the mayor elected in 2008, measured before the quota reform. Every column is weighted by 2006 population and includes state fixed effects. The quota shock is a rate per 10,000 residents using fixed 2006 population and is divided by its main-sample analysis-weighted SD. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. A † attached to a standard error denotes a wild-cluster $p < 0.05$, with resampling by state.

Table 23: Alternative Exclusion Channels

	Alternative Exclusion Channels	
	(1)	(2)
Panel A: Δ Fundo de Participação dos Municípios (FPM) Transfers		
ΔQuota_m	0.082	0.107
	(0.058)	(0.020)†
Panel B: Δ Per-Family Bolsa Família Benefit Value		
ΔQuota_m	0.049	0.083
	(0.043)	(0.035)
Incumbent Party	✓	✓
Bolsa Família Intensity	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors / Municipalities	2,078	2,078
Observations	4,156	4,156

Notes: Panel A uses the change between the 12-month average of total municipal FPM transfers in 2008 and 2012, per 10,000 residents using fixed 2006 population, divided by its full-sample 2006-population-weighted SD. Panel B uses the change between the simple averages of the twelve monthly per-family benefit values in 2008 and 2012, divided by the cross-municipality SD of its change. The analysis-weighted mean changes are R$57,959.64 per month per 10,000 residents for FPM and R$49.48 per family per month for benefit value. Multiplying the standardized Column (2) quota coefficients by the corresponding outcome SD gives R$5,466.59 per month per 10,000 residents and R$0.91 per family per month, respectively. All models use 2006-population weights. Each displayed row is a reduced-form channel diagnostic and is not conditioned on realized beneficiary expansion. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. A † attached to a standard error denotes a wild-cluster $p < 0.05$, with resampling by state.

Table 24: Reverse Causality Channels

	Reverse Causality Channels		
	(1)	(2)	(3)
Panel A: Δ Permitted Bolsa Família Expansion			
Incumbent's Vote Share in 2008	0.009	-0.005	-0.008
	(0.018)	(0.016)	(0.019)
Panel B: Δ Realized Bolsa Família Expansion			
Incumbent's Vote Share in 2008	-0.058	-0.049	-0.042
	(0.026)†	(0.022)	(0.019)†
Incumbent Party	✓	✓	✓
Bolsa Família Intensity	✓	✓	✓
Municipality Characteristics		✓	✓
Mayor Characteristics		✓	✓
Benefit Value			✓
FPM Transfers			✓
Number of Mayors / Municipalities	2,078	2,078	2,078
Observations	2,078	2,078	2,078

Notes: Panel A measures permitted quota expansion in quota-shock-SD units. Panel B measures the 2008–2012 net change in the 12-month average beneficiary-family stock in treatment-change-SD units. The regressor is the 2008 winner's vote share divided by its analysis-weighted SD of 9.59 percentage points. Every column is weighted by 2006 population and includes state fixed effects. Columns (1)–(2) follow the pre-treatment control sequence. Column (3) additionally includes 2008 benefit value and FPM transfers. Standard errors clustered by state are reported in parentheses. A † attached to a standard error denotes a wild-cluster $p < 0.05$, with resampling by state.

Table 25: Monotonicity by Quota-Shock Size

	ΔBeneficiaries$_m$		
	(1)	(2)	(3)
Panel A: Top 50%			
ΔQuota$_m$	0.504	0.545	0.527
	(0.062)†	(0.057)†	(0.056)†
Panel B: Top 25%			
ΔQuota$_m$	0.485	0.518	0.531
	(0.091)†	(0.071)†	(0.066)†
Panel C: Top 15%			
ΔQuota$_m$	0.331	0.440	0.494
	(0.153)	(0.116)†	(0.099)†
Panel D: Top 5%			
ΔQuota$_m$	0.510	0.604	0.630
	(0.181)	(0.124)†	(0.104)†
Panel E: Top 1%			
ΔQuota$_m$	0.776	0.801	0.828
	(0.163)	(0.135)	(0.128)†
Incumbent Party	✓	✓	✓
Bolsa Família Intensity	✓	✓	✓
Municipality Characteristics		✓	✓
Mayor Characteristics		✓	✓
Benefit Value			✓
FPM Transfers			✓

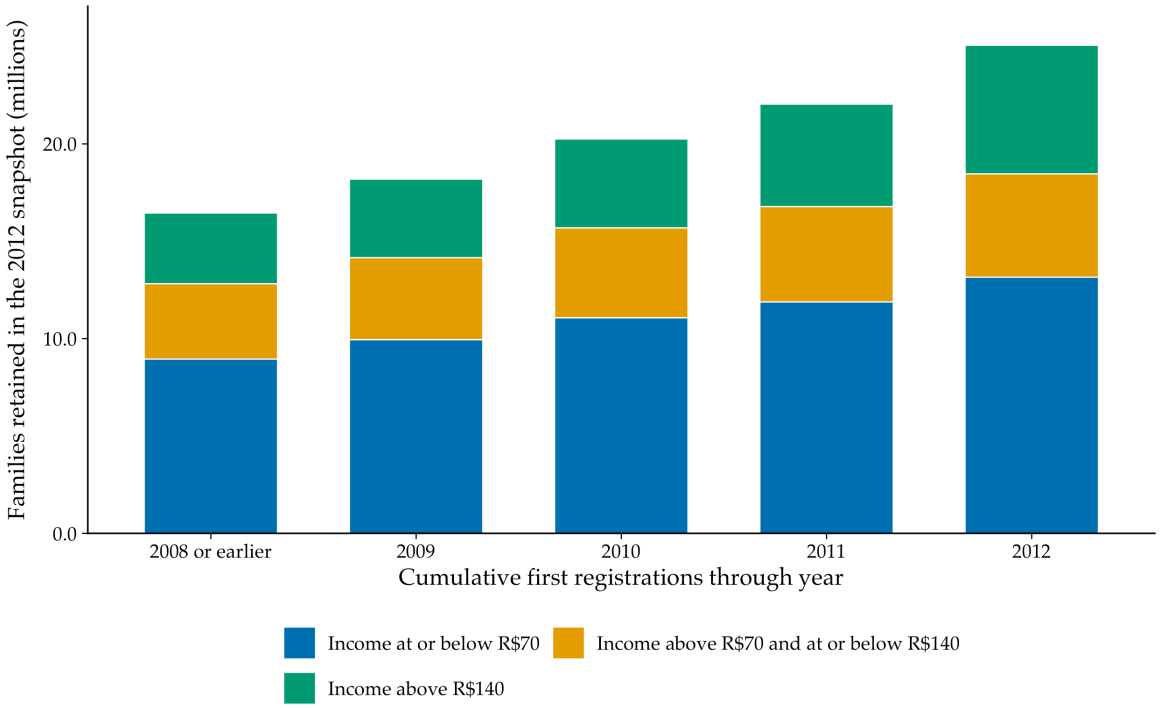
Notes: Each panel estimates separate slopes for the named upper-tail group and its complement on the full locked sample; the table reports the upper-tail slope. Groups are defined by unweighted quota-shock quantiles and are nested. Columns follow the headline control sequence, including baseline benefit value and FPM transfers in Column (3), with 2006-population weights and state-clustered standard errors. † denotes the wild-cluster result. This is an average first-stage-shape diagnostic, not a test of unit-level monotonicity.

Table 26: Family Income in 2012 by First-Registration Cohort

	Registered before 2009		Registered in 2009–2012	
	Mean	SD	Mean	SD
Panel A: Monthly per-capita family income indicators				
Income at or below R$70	0.544	0.498	0.489	0.500
Income above R$70 through R$140	0.235	0.424	0.167	0.373
Income above R$140	0.221	0.415	0.345	0.475
Missing income	0.000	0.000	0.000	0.000
Panel B: Cohort size				
Families	16,452,008		8,612,498	

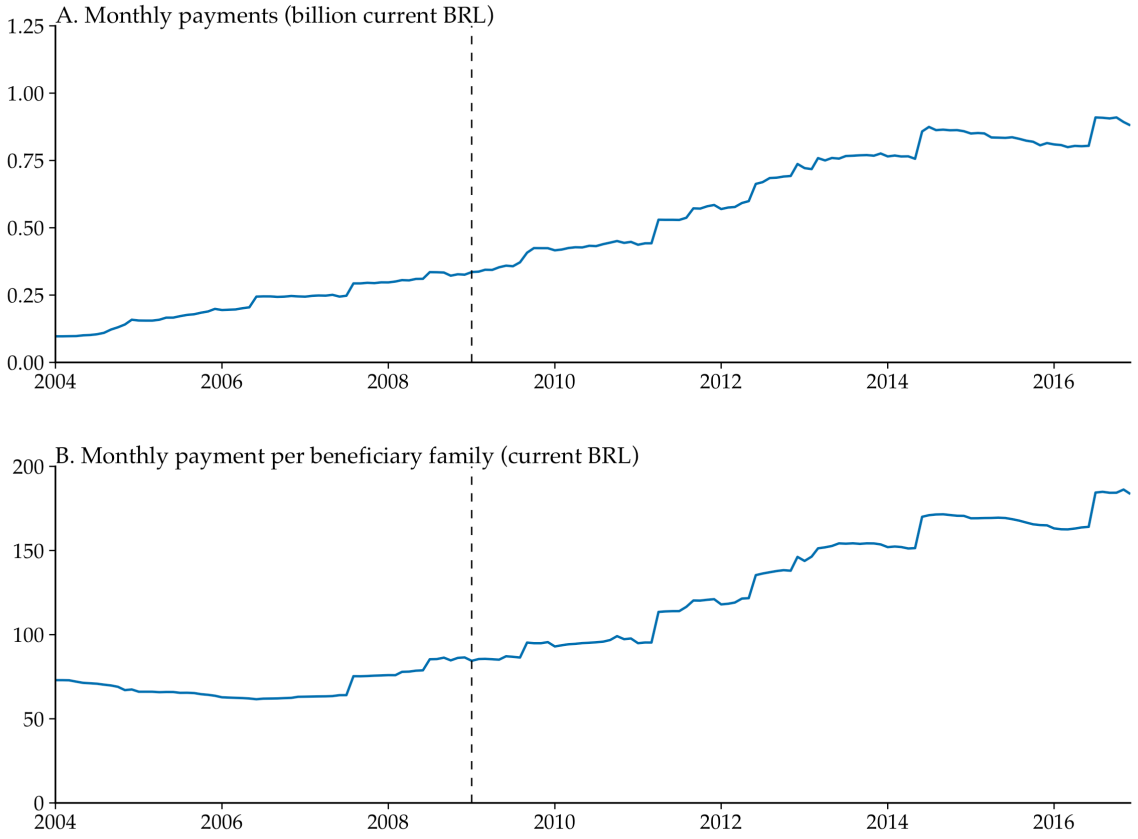
Notes: National, unweighted family-level composition among registrations retained in the 2012 *Cadastro Único* snapshot with valid first-registration dates. All income measures refer to 2012; the first column group is not a 2009 income snapshot. Means are shares, and SD is the sample standard deviation of the corresponding binary indicator. Income groups are mutually exclusive; zero reported income is included in the lowest band. Cohorts include all registered families and do not identify actual *Bolsa Família* receipt. The source includes two unresolved municipality tokens retained in national totals. Invalid registration dates are excluded as documented in the source audit. Individual characteristics, mean income, and household composition cannot be recovered from these grouped counts.

Figure 4: Cumulative Family Registrations Observed in the 2012 *Cadastro Único* Snapshot, by Reported Income



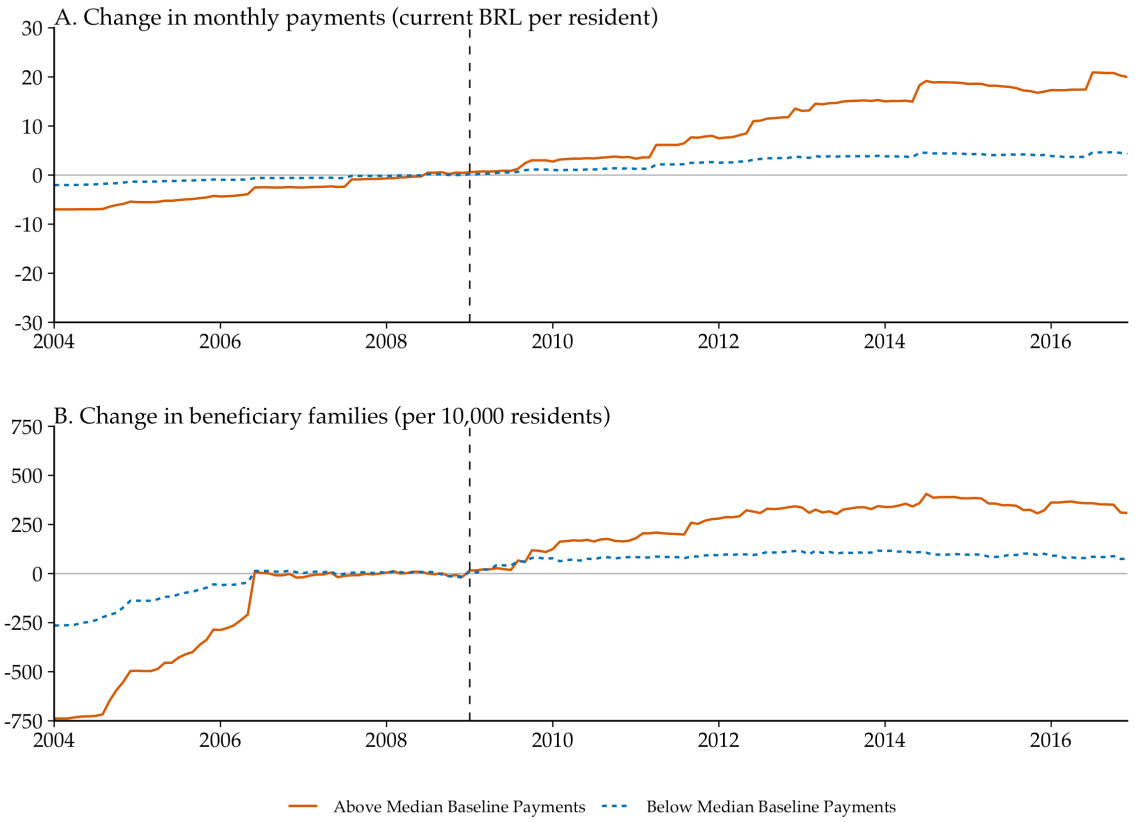
Notes: Counts sum municipality-level family records in the 2012 snapshot. "2008 or earlier" pools all valid registration dates through December 2008. Municipality and nominal monthly per-capita family income are recorded in 2012; income is not measured at registration. Registration dates outside 1900–2012 are excluded. Each bar cumulates families through the indicated registration year, conditional on their presence in the 2012 snapshot. Stacked segments are mutually exclusive 2012 income categories: at most R$70, above R$70 through R$140, above R$140, and missing or invalid income. These are retrospective cohorts, not historical registry stocks. The available aggregates pool years through 2008 and contain no registrations after 2012; they cannot support separate 2004–2008 bars or a series through 2016.

Figure 5: Monthly *Bolsa Família* Payments, 2004–2016



Notes: Totals cover the same 2,078 headline municipalities throughout, rather than Brazil as a whole. Payments are nominal BRL; the mean benefit is aggregate payments divided by beneficiary families. The series retains the existing data-construction corrections for isolated payment spikes. The dashed line marks the year 2009, not the implementation month. Source: MDS monthly municipal program records, distinct from the 2012 family-registration snapshot.

Figure 6: Program Expansion by Baseline Payment Intensity



Notes: Municipalities are divided at the unweighted median of 2008 monthly payments per 2006 resident (BRL 6.55). Within each group, payments and beneficiary-family counts are divided by the group's fixed 2006 population. Each series subtracts its own 2008 calendar-year mean. Monetary values are nominal. These profiles describe heterogeneity in program expansion; they do not estimate the causal effect of baseline payment intensity or an electoral return per real. Source: MDS municipal program records.

Table 27: Quota Expansion and Family Registration

	New registrations per 10,000 residents		
	(1)	(2)	(3)
Panel A: All registered families			
ΔQuota_m (π_1)	29.200	38.371	40.187
	(9.554)	(9.955)	(9.786)
Kleibergen–Paap F	9.34	14.86	16.87
Panel B: Income above R$140			
ΔQuota_m (π_1)	-4.131	-1.589	4.163
	(4.545)	(3.976)	(3.625)
Kleibergen–Paap F	0.83	0.16	1.32
Panel C: Income at or below R$140			
ΔQuota_m (π_1)	33.331	39.960	36.024
	(8.558)	(8.800)	(8.146)
Kleibergen–Paap F	15.17	20.62	19.55
Panel D: Income at or below R$70			
ΔQuota_m (π_1)	30.449	35.883	30.017
	(7.547)	(7.592)	(6.624)
Kleibergen–Paap F	16.28	22.34	20.53
Incumbent Party	✓	✓	✓
Baseline *Bolsa Família* Intensity	✓	✓	✓
Municipality Characteristics		✓	✓
Mayor Characteristics		✓	✓
Benefit Value			✓
FPM Transfers			✓
Municipalities	2,078	2,078	2,078
Observations	4,156	4,156	4,156

Notes: All columns retain the locked 2008–2012 sample, quota-shock instrument, municipality and year fixed effects, 2006-population weights and state-clustered CRV1 standard errors. Controls are interacted with post. Registration expansion counts families first registered in 2009–2012 and still observed in the 2012 *Cadastro Único* snapshot. Income (monthly per capita) and municipality are measured in 2012. These are surviving entry cohorts, not historical net stocks or verified *Bolsa Família* recipients. The income definitions overlap: the at-or-below-R$140 group combines the two lower-income bands. Each coefficient reports the change in family registrations per 10,000 residents (fixed 2006 population) induced by one weighted SD of the quota shock. Natural-rate first stages add across disjoint income groups. Parentheses report standard errors.

Table 28: Quota Effects on Registration and Access by CRAS Presence

	Without (1)	With (2)	Difference (3)
Panel A: All Registrations			
ΔQuota_m (ρ_1)	45.367	37.932	-7.435
	(10.695)	(11.644)	(13.950)
Holm-adjusted p	0.003	0.006	1.000
Panel B: Registrations: Income Above R$140			
ΔQuota_m (ρ_1)	2.372	3.124	0.752
	(5.854)	(3.896)	(5.607)
Holm-adjusted p	0.718	0.439	1.000
Panel C: Registrations: Income at or Below R$140			
ΔQuota_m (ρ_1)	42.995	34.808	-8.187
	(6.473)	(9.916)	(10.842)
Holm-adjusted p	0.000	0.001	1.000
Panel D: Registrations: Income at or Below R$70			
ΔQuota_m (ρ_1)	36.452	29.031	-7.421
	(5.560)	(7.798)	(7.838)
Holm-adjusted p	0.001	0.001	1.000
Panel E: Beneficiary Families			
ΔQuota_m (ρ_1)	58.878	54.639	-4.239
	(7.206)	(10.916)	(10.212)
Holm-adjusted p	0.000	0.000	1.000
Panel F: Monthly Payments per Resident			
ΔQuota_m (ρ_1)	0.920	0.887	-0.033
	(0.119)	(0.172)	(0.165)
Holm-adjusted p	0.000	0.000	1.000
Incumbent Party	✓	✓	✓
Baseline *Bolsa Família* Intensity	✓	✓	✓
Municipality Characteristics	✓	✓	✓
Mayor Characteristics	✓	✓	✓
Benefit Value	✓	✓	✓
FPM Transfers	✓	✓	✓
Municipalities	684	1394	2078

Notes: Coefficients report responses to one fixed headline-sample weighted SD of quota exposure. Registration and beneficiary-family quantities are per 10,000 fixed 2006 residents; payments are nominal monthly BRL per resident. Registrations are surviving cohorts observed in 2012, classified by income in that snapshot. Full controls and trends vary by group. Parentheses report state-clustered SEs; Holm adjustment covers the six outcomes within each column, using 9,999 wild-cluster draws. Difference is With minus Without. CRAS is measured before the reform.

Table 29: Quota Effects on Registration and Access by Active Outreach

	Without (1)	With (2)	Difference (3)
Panel A: All Registrations			
ΔQuota_m (ρ_1)	63.992	33.290	-30.702
	(23.464)	(8.396)	(20.651)
Holm-adjusted p	0.027	0.006	1.000
Panel B: Registrations: Income Above R$140			
ΔQuota_m (ρ_1)	2.996	3.635	0.638
	(4.895)	(4.474)	(5.746)
Holm-adjusted p	0.595	0.428	1.000
Panel C: Registrations: Income at or Below R$140			
ΔQuota_m (ρ_1)	60.996	29.655	-31.341
	(20.125)	(6.372)	(17.075)
Holm-adjusted p	0.006	0.002	1.000
Panel D: Registrations: Income at or Below R$70			
ΔQuota_m (ρ_1)	49.050	25.073	-23.977
	(15.663)	(5.192)	(13.028)
Holm-adjusted p	0.008	0.002	1.000
Panel E: Beneficiary Families			
ΔQuota_m (ρ_1)	79.014	49.968	-29.046
	(19.614)	(7.608)	(15.287)
Holm-adjusted p	0.001	0.001	1.000
Panel F: Monthly Payments per Resident			
ΔQuota_m (ρ_1)	1.249	0.813	-0.436
	(0.301)	(0.126)	(0.232)
Holm-adjusted p	0.001	0.000	1.000
Incumbent Party	✓	✓	✓
Baseline *Bolsa Família* Intensity	✓	✓	✓
Municipality Characteristics	✓	✓	✓
Mayor Characteristics	✓	✓	✓
Benefit Value	✓	✓	✓
FPM Transfers	✓	✓	✓
Municipalities	377	1017	1394

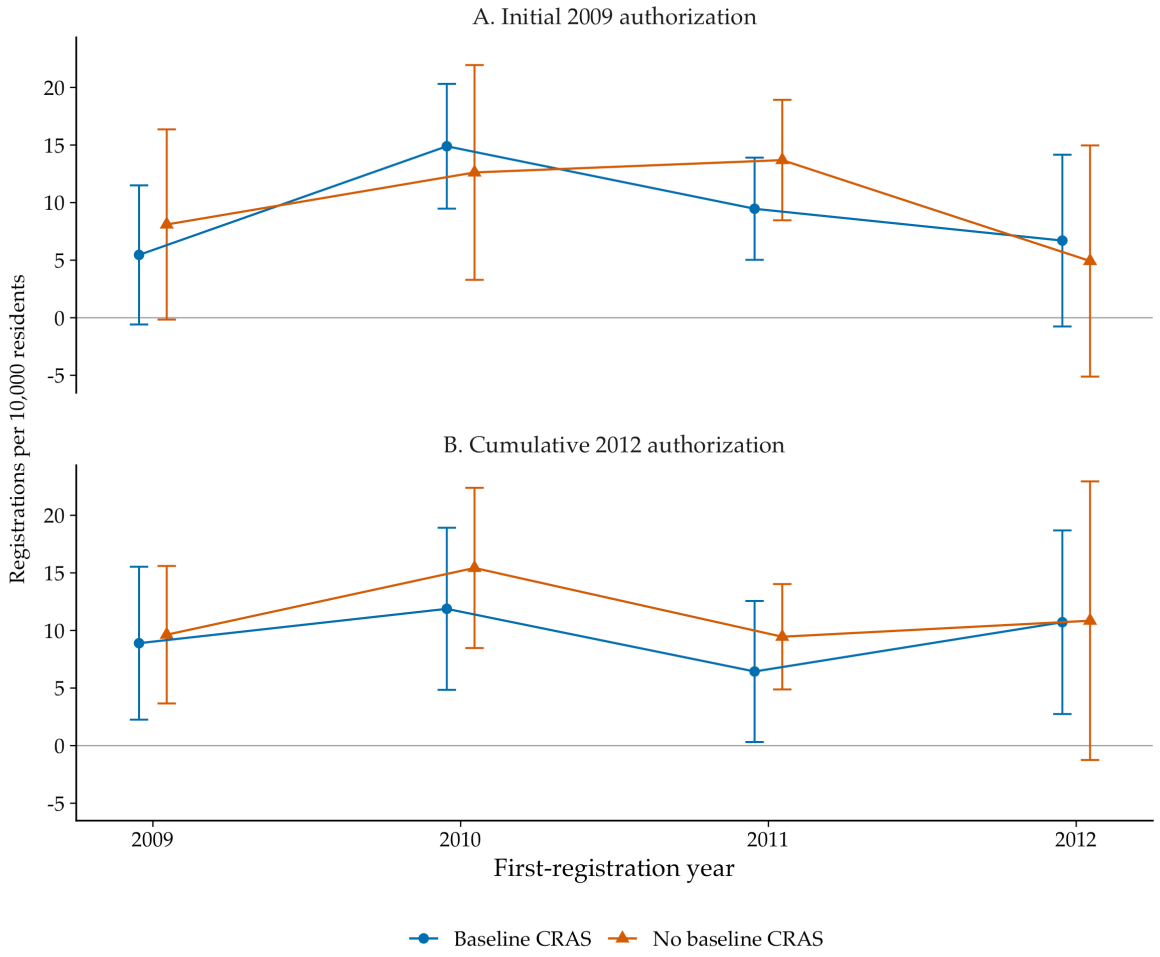
Notes: Coefficients report responses to one fixed headline-sample weighted SD of quota exposure. Registration and beneficiary-family quantities are per 10,000 fixed 2006 residents; payments are nominal monthly BRL per resident. Registrations are surviving cohorts observed in 2012, classified by income in that snapshot. Full controls and trends vary by group. Parentheses report state-clustered SEs; Holm adjustment covers the six outcomes within each column, using 9,999 wild-cluster draws. Difference is With minus Without. Outreach uses verified yes/no baseline reports within CRAS municipalities.

Table 30: Quota Effects on Registration and Access by PAIF Family Follow-Up

	Without (1)	With (2)	Difference (3)
Panel A: All Registrations			
ΔQuota_m (ρ_1)	44.571	36.726	-7.845
	(21.781)	(9.516)	(19.052)
Holm-adjusted p	0.068	0.003	1.000
Panel B: Registrations: Income Above R$140			
ΔQuota_m (ρ_1)	-0.631	4.653	5.284
	(5.043)	(4.453)	(5.652)
Holm-adjusted p	0.913	0.309	1.000
Panel C: Registrations: Income at or Below R$140			
ΔQuota_m (ρ_1)	45.202	32.073	-13.129
	(20.564)	(7.649)	(17.703)
Holm-adjusted p	0.038	0.001	1.000
Panel D: Registrations: Income at or Below R$70			
ΔQuota_m (ρ_1)	33.791	27.672	-6.120
	(17.906)	(6.116)	(15.981)
Holm-adjusted p	0.111	0.001	1.000
Panel E: Beneficiary Families			
ΔQuota_m (ρ_1)	67.393	53.045	-14.348
	(23.567)	(8.601)	(20.442)
Holm-adjusted p	0.022	0.000	1.000
Panel F: Monthly Payments per Resident			
ΔQuota_m (ρ_1)	1.046	0.868	-0.178
	(0.369)	(0.136)	(0.324)
Holm-adjusted p	0.008	0.001	1.000
Incumbent Party	✓	✓	✓
Baseline *Bolsa Família* Intensity	✓	✓	✓
Municipality Characteristics	✓	✓	✓
Mayor Characteristics	✓	✓	✓
Benefit Value	✓	✓	✓
FPM Transfers	✓	✓	✓
Municipalities	432	962	1394

Notes: Coefficients report responses to one fixed headline-sample weighted SD of quota exposure. Registration and beneficiary-family quantities are per 10,000 fixed 2006 residents; payments are nominal monthly BRL per resident. Registrations are surviving cohorts observed in 2012, classified by income in that snapshot. Full controls and trends vary by group. Parentheses report state-clustered SEs; Holm adjustment covers the six outcomes within each column, using 9,999 wild-cluster draws. Difference is With minus Without. PAIF is the existing baseline recorded-family indicator within CRAS municipalities; missing counts retain the canonical zero coding. This proxy is not a verified contact-frequency measure.

Figure 7: Annual Registration Responses by Baseline CRAS



Notes: Each point is the quota-related response of the indicated registration cohort, per 10,000 residents, for municipalities with or without a CRAS before the reform. The two panels use separate authorization measures, each scaled by its own full-sample population-weighted standard deviation. Bars are pointwise state-clustered 95% confidence intervals. Table F.19 reports the group contrasts and multiplicity-adjusted bootstrap tests. Families are observed retrospectively in the 2012 snapshot. Panel B decomposes the cumulative main response by registration year; it does not imply that a future quota change caused earlier registrations. The available aggregates cannot identify pre-2009 monthly trends or families that exited the registry.

Table 31: Electoral Returns and Household Transfers: Conditional Registration Scenarios

	Estimated	No Quota-Induced Registration: Assumed Share			
	Quota Response	0%	25%	50%	100%
	(1)	(2)	(3)	(4)	(5)
Panel A: Family Registrations					
Response / gain forgone	40.19	40.19	40.19	40.19	40.19
	(9.79)				
95% confidence interval	[20.03, 60.34]	[20.03, 60.34]	[20.03, 60.34]	[20.03, 60.34]	[20.03, 60.34]
Holm-adjusted p	0.002	—	—	—	—
Panel B: Beneficiary Families					
Response / gain forgone	54.68	0.00	13.67	27.34	54.68
	(9.48)				
95% confidence interval	[35.16, 74.19]	[0.00, 0.00]	[8.79, 18.55]	[17.58, 37.10]	[35.16, 74.19]
Holm-adjusted p	<0.001	—	—	—	—
Panel C: Monthly Transfers (BRL)					
Response / gain forgone	8,899.60	0.00	2,224.90	4,449.80	8,899.60
	(1,477.41)				
95% confidence interval	[5,856.81, 11,942.38]	[0.00, 0.00]	[1,464.20, 2,985.59]	[2,928.41, 5,971.19]	[5,856.81, 11,942.38]
Holm-adjusted p	<0.001	—	—	—	—
Panel D: Incumbent Vote Share					
Response / gain forgone	1.41	0.00	0.35	0.71	1.41
	(0.48)				
95% confidence interval	[0.42, 2.41]	[0.00, 0.00]	[0.11, 0.60]	[0.21, 1.20]	[0.42, 2.41]
Holm-adjusted p	0.035	—	—	—	—
Municipalities	2,078	—	—	—	—
Observations	4,156	—	—	—	—

Notes: Column (1) estimates responses to 171.50 additional quota slots per 10,000 residents, using full baseline controls, municipality/year effects, 2006-population weights and state-clustered standard errors (parentheses). Intervals are pointwise 95% intervals; reported p-values use 9,999 wild-cluster draws and Holm adjustment across four outcomes. Registrations are surviving 2009–2012 entry cohorts in the 2012 snapshot; beneficiary counts and monthly payments use annual averages. Panels A–C are per 10,000 residents; payments are nominal BRL and votes are percentage points. Columns (2)–(5) suppress all quota-induced registrations (Panel A). For each other outcome, the assumed share multiplies column (1); matching shares across outcomes are illustrative, not estimated. Zero permits gains through other channels, including previously registered families; 100% assumes all gains depend on new registrations. Conditional intervals treat these shares as fixed. These scenarios do not identify mediation or the effect of removing political incentives. Transfers are municipal totals, not payments linked to new registrants, total household income, cumulative transfers or welfare.

Table 32: Electoral Returns Across Control Specifications

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Panel A: First Stage															
Δ Quota (π_1)	0.30	0.49	0.43	0.60	0.56	0.59	0.34	0.54	0.30	0.48	0.30	0.32	0.36	0.37	0.51
	(0.05)	(0.04)	(0.05)	(0.03)	(0.03)	(0.04)	(0.05)	(0.03)	(0.05)	(0.04)	(0.05)	(0.05)	(0.05)	(0.05)	(0.04)
First-Stage F	33.3	138.3	78.8	338.0	278.8	271.5	47.4	237.8	32.8	135.9	34.2	38.3	55.1	61.1	180.2
Panel B: Reduced Form															
Δ Quota (ρ_1)	1.41	0.73	1.57	0.68	0.20	0.42	1.06	0.41	1.48	0.78	1.41	1.16	0.96	0.76	0.64
	(0.48)	(0.34)	(0.66)	(0.28)	(0.44)	(0.37)	(0.39)	(0.46)	(0.49)	(0.33)	(0.48)	(0.48)	(0.36)	(0.34)	(0.35)
Wild-Cluster p	0.028	0.054	0.044	0.028	0.694	0.324	0.038	0.452	0.025	0.033	0.024	0.047	0.041	0.069	0.111
Panel C: Two-Stage Least Squares															
$\Delta\widehat{\text{Beneficiaries}}$ (β_1)	4.69	1.50	3.70	1.14	0.36	0.71	3.11	0.76	4.98	1.61	4.65	3.64	2.67	2.05	1.25
	(1.99)	(0.70)	(1.67)	(0.47)	(0.78)	(0.62)	(1.23)	(0.86)	(2.09)	(0.69)	(1.96)	(1.73)	(1.02)	(0.88)	(0.68)
	[1.242,	[0.072,	[0.471,	[0.156,	[-1.320,	[-0.595,	[0.766,	[-1.027,	[1.384,	[0.202,	[1.256,	[0.518,	[0.624,	[0.186,	[-0.182,
AR 95% Confidence Set	10.282]	2.990]	7.579]	2.096]	1.925]	1.990]	6.131]	2.546]	10.876]	3.092]	10.124]	8.191]	5.020]	3.939]	2.659]
Full Baseline Controls	✓	✓	✓		✓	✓	✓	✓			✓	✓	✓	✓	✓
Baseline Controls Except FPM									✓	✓					
State × Post		✓		✓		✓				✓					✓
Region × Post			✓												
Log 2006 Quota × Post				✓	✓	✓	✓								
Expected Quota Change × Post				✓	✓	✓		✓							
Quadratic Baseline FPM											✓				
Quadratic Log Population												✓		✓	✓
Quadratic Baseline BF Intensity													✓	✓	✓
Municipalities	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078

Notes: All columns use the same 2008/2012 panel, municipality and year effects, 2006-population weights and state-clustered standard errors (parentheses; 26 states). Quota and beneficiary changes use the fixed headline-sample weighted SDs of rates per 10,000 residents. Panels B/C concern incumbent vote share in percentage points. Column (1) is the restored main specification. Column (4) adapts Gerard–Naritomi–Silva's log 2006 quota and predicted quota-change controls with state-period effects; it retains our sample, instrument, horizon and inference and is not a replication of their estimator. All added baseline variables and their quadratic terms are interacted with post. The FPM control is its 2008 level, not post-reform transfers. Wild p-values use 9,999 draws. AR sets use quadratic inversion; disconnected two-tail sets, if present, are displayed by their conservative envelope. The additional diagnostic battery reports 2006 presidential-vote controls separately; no 2010 vote is substituted.

Table 33: Identification Questions and Diagnostic Evidence

	Object of Interest (1)	Evidence Examined (2)	Remaining Requirement (3)
Continuous quota exposure	Electoral reduced form per fixed exposure SD	Support, influence and adjusted response profiles	Conditional parallel trends and dose comparisons remain assumptions
Binary quota contrasts	Electoral contrast across a declared cutoff	Zero, median and upper-quartile quota instruments	Changing the cutoff changes the comparison and potentially its weights
Beneficiary-expansion IV	Electoral response per induced enrollment SD	First stage, reduced form, 2SLS and AR sets	Relevance alone does not establish exclusion or monotonicity
Initial versus cumulative quota	Same cumulative enrollment and election outcomes	Two instruments under matched control specifications	Federal authorization does not establish exogeneity of later revisions
Predetermined political controls	Same estimand with alternative baseline adjustments	2006 PT vote, state-period and allocation controls	Stability is informative but cannot certify absence of confounding
Fiscal allocation and exclusion	Quota–FPM association and direct-effect scenarios	Historical-rule audit and exclusion sensitivity	Shared formula inputs may explain correlation; residual channels can remain
Registration and local contact	Registration responses and conditional subgroup contrasts	Income groups and predetermined CRAS/contact measures	Subgroup heterogeneity does not identify causal mediation or office placement
Weak-IV and selection sensitivity	Uncertainty and stability under explicit departures	Quadratic AR inversion, wild RF tests and existing Oster	Robust inference does not repair an invalid instrument

Notes: This matrix organizes additional diagnostic evidence without changing the headline specification. The principal horizon is the 2008–2012 electoral comparison and cumulative program expansion. SD denotes the fixed headline-sample, population-weighted standard deviation. Binary quota instruments are distinct from binary beneficiary treatments. Oster sensitivity concerns reduced-form coefficient stability; it is not an IV exclusion test.

Table 34: Support and Influence Under the Frozen Headline Sample

Diagnostic	Value
	(1)
Municipalities	2,078
States	26
Population-weighted quota-rate SD	171.496
Within-state share of weighted quota variance	0.713
Residual variance retained after GNS allocation controls	0.878
Largest municipality partial-leverage share	0.045
Top-ten municipality partial-leverage share	0.158
Largest state partial-leverage share	0.148 (BA)

Notes: Quota exposure is the cumulative 2008–2012 authorization shock per 10,000 residents. Residual support partials out the full predetermined headline controls, incumbent-party indicators, and state indicators using 2006-population weights. The GNS row then adds log 2006 quota and the predicted quota change to measure how much residual exposure remains after partialling out their mechanical allocation component. Partial leverage is each municipality's weighted squared residual quota exposure divided by its sample sum; it is diagnostic and does not justify deleting influential observations.

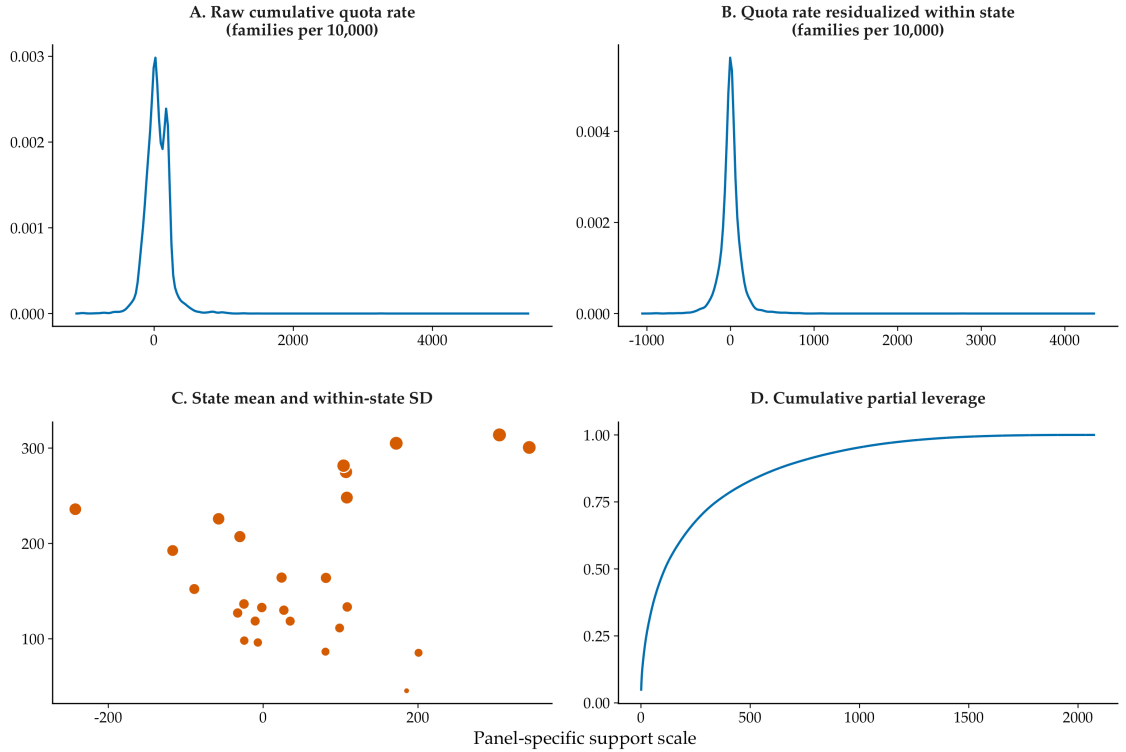


Figure 8: Quota Support, Within-State Variation, and Partial Leverage

Notes: The locked headline sample has 2,078 municipalities and uses 2006-population weights. Panel B partials out the full predetermined headline controls, incumbent-party indicators, and state indicators. Partial leverage is proportional to weighted squared residual quota exposure. The figure diagnoses concentration and does not define an observation-deletion rule.

Table 35: Continuous and Binary Quota Instruments: Frozen Headline Contract

	Continuous (1)	Positive (2)	National P50 (3)	State P50 (4)	National P75 (5)	State P75 (6)
Panel A: First Stage						
Quota Exposure (π_1)	0.301	0.215	0.306	0.322	0.522	0.291
	(0.052)	(0.090)	(0.080)	(0.082)	(0.084)	(0.159)
First-Stage F	33.3	5.8	14.8	15.6	38.5	3.4
Panel B: Reduced Form						
Quota Exposure (ρ_1)	1.414	4.359	4.152	2.835	2.430	3.195
	(0.482)	(1.361)	(1.544)	(1.411)	(2.108)	(0.909)
Wild-Cluster p	0.023	0.027	0.049	0.133	0.345	0.018
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}$ (β_1)	4.690	20.244	13.556	8.804	4.652	10.967
	(1.991)	(10.062)	(6.023)	(5.039)	(3.888)	(5.459)
AR 95% Confidence Set	[1.242, 10.282]	[6.545, 138.379]	[3.113, 34.191]	[-0.216, 25.217]	[-4.154, 12.832]	$[-\infty, +\infty]$
Municipalities	2,078	2,078	2,078	2,078	2,078	2,078

Notes: All columns use the locked 2,078-municipality sample, the same cumulative 2008–2012 treatment scaled by its fixed headline SD (181.365 families per 10,000), full predetermined controls, 2006-population weights, municipality and year effects, and state clustering. The continuous instrument uses its fixed headline SD; binary first stages are one-versus-zero contrasts. Cutoffs were fixed on the full available 2008 municipal universe in regressions.csv before estimation, using unweighted quantiles with ties included. The positive comparison assigns zero to the reference group. Wild p-values use 9,999 state-cluster draws; AR sets use the t(25) critical value. Exact disconnected components and natural-unit coefficients are retained in sidecars. Panel B concerns incumbent vote share in percentage points; Panel C reports percentage points per fixed headline SD of beneficiary expansion.

Table 36: Continuous and Binary Quota Instruments: State-by-Period Sensitivity

	Continuous (1)	Positive (2)	National P50 (3)	State P50 (4)	National P75 (5)	State P75 (6)
Panel A: First Stage						
Quota Exposure (π_1)	0.487	0.493	0.615	0.571	0.713	0.560
	(0.041)	(0.103)	(0.102)	(0.087)	(0.136)	(0.157)
First-Stage F	138.3	22.9	36.3	43.4	27.5	12.8
Panel B: Reduced Form						
Quota Exposure (ρ_1)	0.731	2.519	0.803	-0.499	0.622	0.500
	(0.338)	(1.294)	(1.361)	(1.087)	(1.466)	(1.003)
Wild-Cluster p	0.057	0.082	0.640	0.680	0.707	0.607
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}$ (β_1)	1.499	5.111	1.305	-0.873	0.873	0.894
	(0.697)	(2.567)	(2.088)	(1.952)	(1.975)	(1.650)
AR 95% Confidence Set	[0.072, 2.990]	[-0.335, 11.406]	[-4.249, 5.076]	[-5.701, 2.836]	[-4.511, 4.506]	[-5.042, 3.805]
Municipalities	2,078	2,078	2,078	2,078	2,078	2,078

Notes: All columns use the locked 2,078-municipality sample, the same cumulative 2008–2012 treatment scaled by its fixed headline SD (181.365 families per 10,000), full predetermined controls, 2006-population weights, municipality and year effects, and state clustering. The continuous instrument uses its fixed headline SD; binary first stages are one-versus-zero contrasts. Cutoffs were fixed on the full available 2008 municipal universe in regressions.csv before estimation, using unweighted quantiles with ties included. The positive comparison assigns zero to the reference group. Wild p-values use 9,999 state-cluster draws; AR sets use the t(25) critical value. Exact disconnected components and natural-unit coefficients are retained in sidecars. Every column additionally absorbs a separate post-period shift for each state. Panel B concerns incumbent vote share in percentage points; Panel C reports percentage points per fixed headline SD of beneficiary expansion.

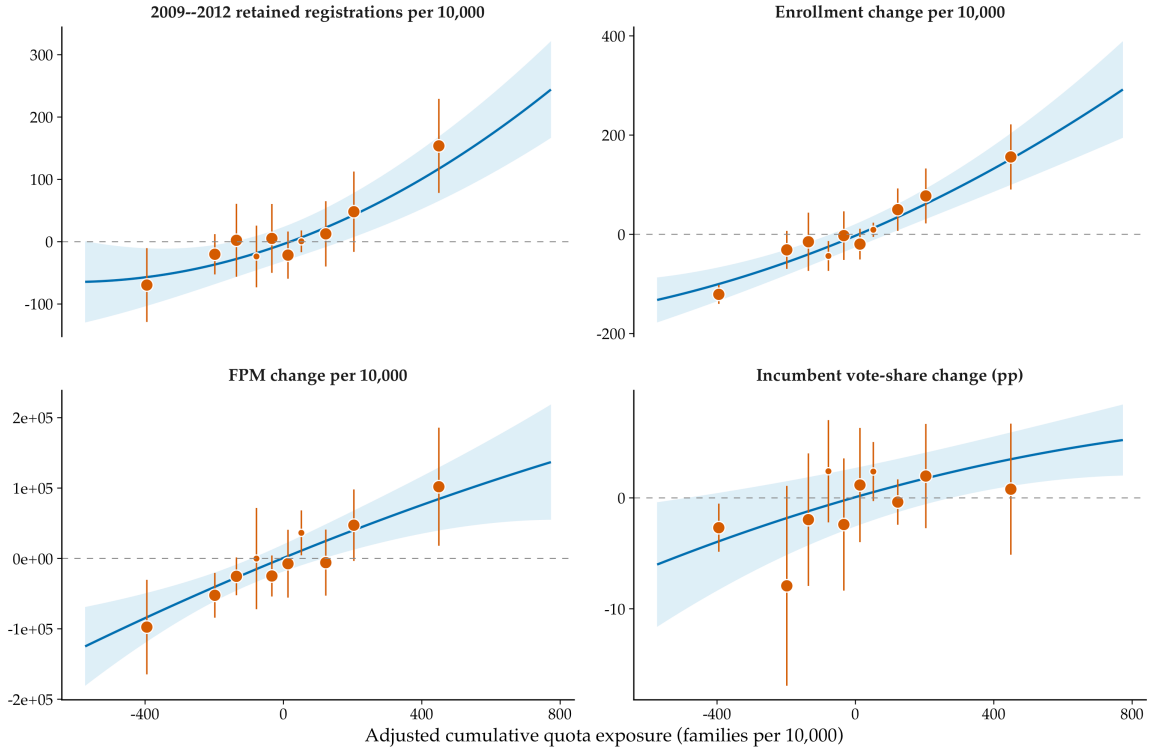


Figure 9: Adjusted Quota-Response Profiles

Notes: Points are fixed decile-bin means with state-clustered 95% intervals; lines are predeclared quadratic summaries over the first through ninety-ninth percentiles of adjusted quota support. Outcomes and quota exposure are residualized on the full predetermined headline controls and incumbent-party indicators using 2006-population weights. Enrollment shows first-stage shape, votes show reduced-form shape, and registration and FPM are channel associations. None of the curves is a causal derivative of endogenous enrollment.

Table 37: Focused Presidential and Allocation Control Comparisons

Specification	Main (1)	+ PT (2)	+ State (3)	+ State and PT (4)	+ GNS (5)	+ GNS, State, and PT (6)
Panel A: First Stage						
Quota shock	0.301	0.312	0.487	0.492	0.559	0.587
	(0.052)	(0.053)	(0.041)	(0.043)	(0.033)	(0.037)
First-stage F	33.3	34.5	138.3	131.4	278.8	256.5
Panel B: Reduced Form						
Quota shock	1.414	1.280	0.731	0.694	0.200	0.464
	(0.482)	(0.418)	(0.338)	(0.346)	(0.440)	(0.370)
Wild-cluster p	0.023	0.019	0.051	0.067	0.687	0.277
Panel C: Two-Stage Least Squares						
Induced beneficiary expansion	4.690	4.106	1.499	1.412	0.358	0.790
	(1.991)	(1.657)	(0.697)	(0.711)	(0.781)	(0.627)
AR 95% confidence set	[1.242, 10.282]	[1.212, 8.696]	[0.072, 2.990]	[-0.037, 2.943]	[-1.320, 1.925]	[-0.513, 2.093]
PT presidential share × Post		✓		✓		✓
State × Post			✓	✓		✓
GNS allocation controls × Post					✓	✓
Municipalities	2,078	2,078	2,078	2,078	2,078	2,078

Notes: All columns use the same matched 2008/2012 headline panel, municipality and year effects, 2006-population weights, and state-clustered standard errors (parentheses). Main includes the full predetermined-control and incumbent-party-trend specification. Quota and beneficiary changes use their fixed headline-sample weighted SDs of rates per 10,000 residents; the outcome is incumbent vote share in percentage points. PT 2006 is the second-round presidential vote share, computed after summing candidate votes across zones; it enters interacted with post. The GNS allocation controls are log 2006 quota and expected quota change. The reduced-form p-value uses 9,999 null-imposed wild-cluster draws. AR sets use quadratic inversion; the exact connected components, including unbounded tails, are retained in the sidecar. The clustered-covariance correction and raw target variances are separately audited for all RF, FS, and outcome-plus-endogenous equations.

Table 38: Initial and Cumulative Instruments: Accepted Annual Endpoint

	Initial / Main (1)	Initial / State-period (2)	Cumulative / Main (3)	Cumulative / State-period (4)
Panel A: First Stage				
Quota Exposure (π_1)	0.255	0.364	0.301	0.487
	(0.037)	(0.050)	(0.052)	(0.041)
First-Stage F	48.1	53.1	33.3	138.3
Panel B: Reduced Form				
Quota Exposure (ρ_1)	0.631	0.060	1.414	0.731
	(0.671)	(0.585)	(0.482)	(0.338)
Wild-Cluster p	0.437	0.928	0.023	0.057
Panel C: Two-Stage Least Squares				
$\Delta\widehat{\text{Beneficiaries}}$ (β_1)	2.474	0.165	4.690	1.499
	(2.570)	(1.601)	(1.991)	(0.697)
AR 95% Confidence Set	[-3.245, 7.846]	[-3.561, 3.335]	[1.242, 10.282]	[0.072, 2.990]
Municipalities	2,078	2,078	2,078	2,078

Notes: Initial and cumulative quota instruments are both divided by the fixed headline instrument SD (171.496) and compared with the same treatment endpoint within the table. Treatment is divided by the fixed headline treatment-change SD (181.365), so 2SLS estimates remain comparable to the headline effect. State-period columns add separate post shifts by state. All columns otherwise use the locked headline sample, controls, weights, fixed effects, clustering, 9,999-draw wild inference, and exact quadratic AR construction. The treatment endpoint is the accepted December-2012 MM12 measure; its January–December window includes months after the October 7 first round. Panel B concerns incumbent vote share in percentage points; Panel C reports percentage points per fixed headline SD of beneficiary expansion.

Table 39: Initial and Cumulative Instruments: September-2012 Treatment Endpoint

	Initial / Main (1)	Initial / State-period (2)	Cumulative / Main (3)	Cumulative / State-period (4)
Panel A: First Stage				
Quota Exposure (π_1)	0.254	0.361	0.285	0.467
	(0.036)	(0.050)	(0.051)	(0.042)
First-Stage F	50.0	52.8	31.1	122.1
Panel B: Reduced Form				
Quota Exposure (ρ_1)	0.631	0.060	1.414	0.731
	(0.671)	(0.585)	(0.482)	(0.338)
Wild-Cluster p	0.437	0.928	0.023	0.057
Panel C: Two-Stage Least Squares				
$\Delta\widehat{\text{Beneficiaries}}$ (β_1)	2.484	0.166	4.961	1.566
	(2.576)	(1.617)	(2.122)	(0.734)
AR 95% Confidence Set	[-3.265, 7.833]	[-3.588, 3.377]	[1.310, 11.018]	[0.075, 3.155]
Municipalities	2,078	2,078	2,078	2,078

Notes: Initial and cumulative quota instruments are both divided by the fixed headline instrument SD (171.496) and compared with the same treatment endpoint within the table. Treatment is divided by the fixed headline treatment-change SD (181.365), so 2SLS estimates remain comparable to the headline effect. State-period columns add separate post shifts by state. All columns otherwise use the locked headline sample, controls, weights, fixed effects, clustering, 9,999-draw wild inference, and exact quadratic AR construction. Only the endogenous treatment endpoint changes: the October-2011 through September-2012 MM12 window ends before the October 7 first round. The cumulative quota instrument itself is not labeled pre-election. Registration cohorts are annual and cannot support an analogous election-date split. Panel B concerns incumbent vote share in percentage points; Panel C reports percentage points per fixed headline SD of beneficiary expansion.

Table 40: FPM Historical-Rule and Measurement Audit

	Verified Evidence (1)	Use in This Battery (2)	Unresolved Limit (3)
Realized FPM transfers	Tesouro Transparente annual CSVs, 2008–2012; cash-basis monthly decendio values from SIAFI	Nominal annual municipality totals and 2008–2012 changes	The CSV and metadata do not label the FPM row gross or net; no price deflation is applied
Matched coverage	5,498 municipalities in the clean FPM panel; all 2,078 headline municipalities complete	Pool adjustments use the complete matched FPM panel, not the rerunner sample	The matched pool is not labeled the Brazilian national total
Ordinary transfer deductions	Official metadata records 18.33 percent Fundeb in 2008 and 20 percent from 2009; the STN guide records 1 percent PASEP	Explains why gross/net reconciliation is necessary	No deduction fields are retained in fpm-panel.csv
December EC 55/2007 parcel	Official STN guide: additional one percent, no Fundeb deduction, PASEP remains	Annual totals mix ordinary and December parcels with different deductions	The clean annual panel does not separate the December parcel
Interior allocation rule	Decree-Law 1,881/1981 brackets 0.6–4.0; 86.4 percent Interior pool; fixed state shares under TCU Resolution 242/1990	Predetermined 2006 population bracket proxy and state-by-bracket-period adjustment	Proxy is not an annual TCU coefficient
Capital and Reserve allocations	Official STN guide: 10 percent Capital and 3.6 percent Reserve; Reserve also receives an Interior share	None without historical classifications and factors	Capital/Reserve formulas, income factors, and annual classifications are absent
Historical coefficients	TCU calculates annual participation percentages; the STN guide identifies DN 92/2008 for exercise 2009	No coefficient reconstruction in this battery	Official 2008–2012 DN annexes, memories, and judicial adjustments have not been recovered
Coefficient-change proxy	377 of 2,078 headline municipalities change statutory bracket when 2010 replaces 2006 population	Documented only; the principal adjustment keeps predetermined 2006 population	The 2010 proxy is potentially post-treatment and is not an actual TCU coefficient

Notes: Checked September 19, 2026 against the official Tesouro Transparente dataset metadata, the March 2023 STN FPM guide, Decree-Law 1,881/1981, EC 55/2007, and the TCU coefficient portal. No current rule is imposed on historical transfers. The 2006 population bracket is only a predetermined proxy. Formula administration does not establish instrument exogeneity.

Table 41: Residual Support After FPM Allocation-Proxy Adjustments

	(A) (1)	(B) (2)	(C) (3)
Adjustment	Headline predetermined controls	Headline controls plus state-period	Headline controls plus state-by-2006-bracket-period proxy
Cells	—	26	272
Singleton cells	0	0	82
Municipalities in singleton cells	0	0	82
Reported municipalities	2,078	2,078	2,078
Municipalities with residual quota support	2,078	2,078	1,996
Weighted SD of signed residual quota	0.450	0.379	0.334
Within-design rank / columns	38 / 38	63 / 63	309 / 309
Collinear terms	None	None	None

Notes: Cells are defined on the 2,078-municipality locked headline sample. Residual support counts municipalities with a post-period quota residual above 1e-10 after the displayed controls and municipality/year effects. The bracket adjustment has 82 singleton cells: those municipalities remain among the 2,078 reported observations but have no residual quota support. The displayed residual spread is the weighted SD of the signed post residual. The schedule comes from Decree-Law 1,881/1981 applied to 2006 population; it does not recover annual TCU coefficients, Capital/Reserve factors, or judicial changes.

Table 42: Quota Exposure and Realized FPM Transfers

	(A) (1)	(B) (2)	(C) (3)	(D) (4)	(E) (5)	(F) (6)
Quota–FPM Reduced Forms						
Quota exposure (FPM-change SD per quota SD)	0.107	0.054	0.079	0.089	0.137	0.019
	(0.020)	(0.012)	(0.016)	(0.013)	(0.023)	(0.010)
Wild-cluster RF p-value	<0.001	<0.001	<0.001	<0.001	<0.001	0.092
Reported municipalities	2,078	2,078	2,078	2,078	2,078	2,078
Municipalities with residual quota support	2,078	2,078	2,078	1,996	1,996	1,996
State clusters	26	26	26	26	26	26

Notes: Every outcome is scaled by the fixed headline-sample, 2006-population-weighted SD of its 2008–2012 change; the instrument is the same fixed one-SD cumulative quota exposure used in the headline design. Columns A and B use realized nominal FPM per 10,000 fixed 2006 residents. A exactly reproduces the accepted channel diagnostic (0.107); B adds the baseline benefit-value and baseline-FPM controls in the full headline nuisance set (0.054). Dividing annual FPM by 12 has no effect after scaling by its change SD and therefore does not explain that difference. C adds state-period effects; D adds state-by-statutory-2006-population-bracket-period proxy effects. E subtracts log growth of the complete 5,498-municipality matched FPM panel, which is not a certified national pool. Because this subtraction is a common year-specific constant absorbed by year effects, E is a log-FPM functional-form check up to rescaling, not an independent removal of aggregate confounding. F uses a municipality's share of its matched state pool and therefore has different outcome units rather than the same estimand. These are descriptive fiscal reduced forms, not causal fiscal effects induced by enrollment. Nominal FPM is not adjusted for inflation.

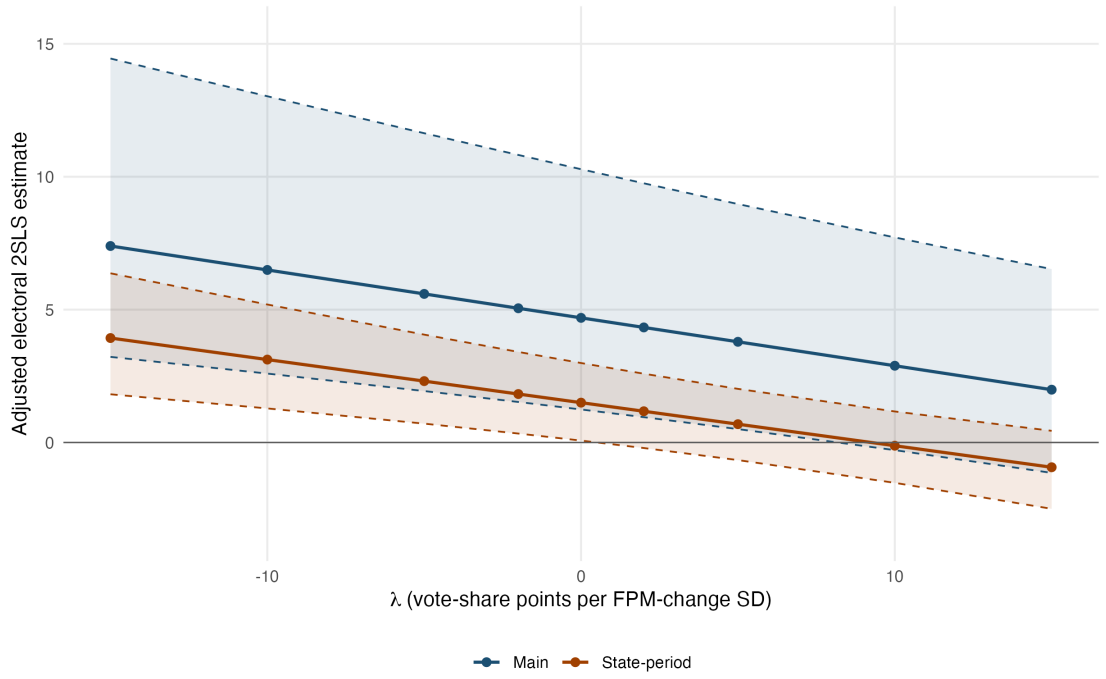


Figure 10: Electoral IV Sensitivity to a Direct FPM Channel

Notes: Solid lines and points are adjusted 2SLS estimates; dashed lines and shaded bands are exact quadratic Anderson–Rubin 95% set limits for each connected component. Main uses the full predetermined headline controls; State-period adds state-by-post effects. λ is an uncalibrated vote-share percentage-point response to one fixed weighted SD of nominal 2008–2012 FPM growth per 10,000 fixed 2006 residents. No external electoral elasticity is used. Upward or downward open triangles at a frame edge mark an unbounded AR limit. The generated grid contains 0 unbounded AR limits.

Table 43: Electoral IV Sensitivity to a Direct FPM Channel

	Assumed Direct RF (1)	Adjusted RF (2)	Adjusted 2SLS (3)	Adjusted AR 95% Set (4)	Adjusted RF Wild p (5)
Panel A: Main					
λ =-15.0	-0.815	2.229	7.393	[3.221, 14.447]	0.004
λ =-10.0	-0.543	1.957	6.492	[2.593, 13.027]	0.005
λ =-5.0	-0.272	1.686	5.591	[1.936, 11.636]	0.011
λ =-2.0	-0.109	1.523	5.050	[1.525, 10.819]	0.018
λ =0.0	0.000	1.414	4.690	[1.242, 10.282]	0.026
λ =2.0	0.109	1.305	4.330	[0.953, 9.752]	0.036
λ =5.0	0.272	1.142	3.789	[0.505, 8.972]	0.059
λ =10.0	0.543	0.871	2.888	[-0.287, 7.716]	0.133
λ =15.0	0.815	0.599	1.987	[-1.143, 6.524]	0.310
Panel B: State-period					
λ =-15.0	-1.185	1.916	3.930	[1.814, 6.366]	0.002
λ =-10.0	-0.790	1.521	3.120	[1.282, 5.192]	0.004
λ =-5.0	-0.395	1.126	2.310	[0.710, 4.059]	0.011
λ =-2.0	-0.158	0.889	1.824	[0.337, 3.408]	0.025
λ =0.0	0.000	0.731	1.499	[0.072, 2.990]	0.052
λ =2.0	0.158	0.573	1.175	[-0.208, 2.588]	0.115
λ =5.0	0.395	0.336	0.689	[-0.661, 2.018]	0.331
λ =10.0	0.790	-0.059	-0.121	[-1.517, 1.168]	0.843
λ =15.0	1.185	-0.454	-0.931	[-2.495, 0.439]	0.203

Notes: λ is the assumed vote-share percentage-point response to one fixed weighted SD of realized FPM-rate growth. That SD is 610,502.8 nominal BRL per 10,000 fixed 2006 residents, or 61.05 nominal BRL per resident; no inflation adjustment is made. Each row estimates $Y - \lambda FPM$ and recomputes the reduced form, 2SLS, exact quadratic AR components, and 9,999-draw state-cluster wild-bootstrap RF test. The assumed direct RF is λ times the specification-specific quota–FPM association. The grid is an uncalibrated sensitivity range; no external electoral elasticity is used, and λ is not a causal FPM estimate. The descriptive point-estimate breakdowns are λ =26.026 (Main) and λ =9.253 (State-period). The first nonnegative evaluated grid values whose AR set includes zero are respectively λ =10.0 and λ =2.0; these are grid values, not exact breakdown thresholds.

Table 44: What the Existing and FPM-Calibrated Sensitivities Establish

	Result (1)	Interpretation (2)
RF proportional-selection stability	Short/long RF = 1.929/1.299; delta* = 2.787 (1.3R2) and 0.697 (2.2R2)	Approximate reduced-form coefficient stability; not an IV exclusion bound
Existing bounded-direct-effect screen	RF = 1.414; strict AR = [1.242, 10.282]; RF breakdown share = 29.8 percent	Mechanical RF-share bounds; not calibrated to a measured fiscal electoral elasticity
Accepted FPM channel diagnostic	Quota–FPM association = 0.107 FPM-change SD per quota SD	Exactly reproduced by Model A; its channel-specification controls omit baseline benefit value and baseline FPM
Full-headline FPM association	Quota–FPM association = 0.054; point breakdown lambda = 26.026	Model B adds the two predetermined baseline controls; annual versus monthly-average scaling cannot explain the difference because SD scaling cancels it
State-period FPM sensitivity	Quota–FPM association = 0.079; point breakdown lambda = 9.253	Joint-covariance sensitivity with declared lambda values; lambda is not estimated

Notes: The Oster sidecar compares reduced-form coefficient stability between a short and a longer predetermined-control specification; it is not an IV exclusion test. The existing bounded-direct-effect sidecar uses shares of the reduced-form point estimate, rather than an externally measured fiscal electoral effect. The new FPM battery supplies a measured quota–FPM association but retains an explicitly uncalibrated electoral-response grid.

Table 45: Focused Registration Responses by Income

Specification	Main (1)	+ PT 2006 (2)	+ State-period (3)	+ GNS (4)
Panel A: Registration responses				
All registrations	40.187	41.231	63.908	73.860
	(9.786)	(9.765)	(8.732)	(7.826)
Holm wild p	0.005	0.004	<0.001	<0.001
Income above R$140	4.163	3.952	7.412	6.557
	(3.625)	(3.670)	(3.681)	(4.275)
Holm wild p	0.538	0.538	0.214	0.412
Income at or below R$140	36.024	37.278	56.496	67.303
	(8.146)	(8.223)	(6.838)	(5.840)
Holm wild p	<0.001	<0.001	<0.001	<0.001
Income at or below R$70	30.017	30.975	46.438	53.402
	(6.624)	(6.808)	(5.746)	(5.613)
Holm wild p	0.001	<0.001	<0.001	<0.001

Notes: Cells report reduced-form responses to one headline-sample weighted SD of the quota shock; parentheses contain state-clustered standard errors. All models use 2,078 municipalities, municipality and year effects, the full headline baseline controls, and 2006-population WLS weights. Registration outcomes are surviving cohorts observed in the 2012 *Cadastro Único* snapshot, classified by 2012 income and expressed per 10,000 fixed 2006 residents. Holm adjusts all 16 outcome-specification tests. Every p-value uses 9,999 null-imposed wild-cluster draws.

Table 46: Focused CRAS, Outreach, and PAIF Response Differences

Specification	Main (1)	+ PT 2006 (2)	+ State-period (3)	+ GNS (4)
Panel B: CRAS minus no CRAS				
All registrations: group difference	-7.435	-7.813	4.967	12.784
	(13.950)	(13.768)	(16.724)	(13.548)
Holm wild p	1.000	1.000	1.000	1.000
Incumbent vote share: group difference	2.239	2.101	1.795	0.899
	(0.736)	(0.739)	(0.785)	(0.817)
Holm wild p	0.317	0.338	0.404	1.000
Panel C: Verified outreach minus no outreach within CRAS				
All registrations: group difference	-30.702	-27.708	-15.452	-3.772
	(20.651)	(18.851)	(22.870)	(21.814)
Holm wild p	1.000	1.000	1.000	1.000
Incumbent vote share: group difference	1.247	1.153	-0.082	0.017
	(0.897)	(1.115)	(1.061)	(1.004)
Holm wild p	1.000	1.000	1.000	1.000
Panel D: Positive minus zero PAIF registry within CRAS				
All registrations: group difference	-7.845	-7.710	3.579	-3.550
	(19.052)	(19.817)	(20.660)	(20.251)
Holm wild p	1.000	1.000	1.000	1.000
Incumbent vote share: group difference	1.598	1.344	-0.617	0.605
	(1.004)	(1.068)	(1.020)	(1.080)
Holm wild p	1.000	1.000	1.000	1.000

Notes: Cells report fully interacted group-one minus group-zero reduced-form responses to one headline-sample weighted SD of the quota shock; parentheses contain state-clustered standard errors. Models use 2,078 municipalities for CRAS and 1,394 for within-CRAS outreach and PAIF, municipality and year effects, full headline baseline controls, and 2006-population WLS weights; group-specific nuisance terms and a group-by-post effect are included. Registration is the surviving 2012 *Cadastro Único* snapshot cohort per 10,000 fixed 2006 residents; vote share is in percentage points. Outreach uses literal verified 2008 Censo SUAS yes/no reports. PAIF (Family Protection and Comprehensive Assistance Service) uses the existing positive family-count indicator; its upstream questionnaire mapping remains provisional. Holm adjusts the eight tests within each CRAS, outreach, and PAIF family. Every p-value uses 9,999 null-imposed wild-cluster draws. These descriptive differences do not identify the effect of installing CRAS, outreach, PAIF, or causal mediation.